

LAUNDER-BENCH: CAN REPOSITORY-LEVEL AGENTIC CODE LAUNDERING BE DETECTED?

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ABSTRACT

Coding agents can now rewrite repository-scale codebases while preserving tested behavior, creating a new challenge for plagiarism detection, license auditing, and software provenance. We ask whether such rewrites can still be traced to their source. Existing clone benchmarks emphasize fragments and bounded edits, while repository-transformation benchmarks evaluate functional correctness rather than provenance. Moreover, ranking the most similar source does not establish derivation when independent implementations can be similarly close. We define *repository-level agentic code laundering* and introduce Launder-Bench, to our knowledge the first repository-level benchmark for provenance under agentic rewriting, constructed from 282 real-world repositories. A source-hidden reconstruction protocol decomposes source behavior, reconstructs modules and repository structure, and retains rewrites only after execution-based verification. We evaluate the same provenance problem in two complementary settings: Launder-Bench-Pair contains 1,128 pairwise instances asking whether two repositories share provenance, while Launder-Bench-Attr contains 229 source-attribution instances asking which candidate is the source, with abstention when evidence is insufficient. Measured same-domain independent repositories establish how much similarity can arise without shared provenance and provide detector-specific reference bands. Existing methods remain far from saturating Launder-Bench: the strongest static method reaches only 0.66 AUROC on pairwise detection, while repository encoders rank the true source first on up to 22% of attribution queries but earn calibrated attribution credit on only 1–2%. These results reveal a systematic gap between relative code similarity and source-specific provenance evidence, providing a reproducible target for provenance detection under agentic rewriting. <https://github.com/HuieL/Launder-Bench>

1 INTRODUCTION

Coding agents can now rewrite repository-scale code while preserving tested behavior. Agentic rewriting can rename identifiers, restructure module boundaries, reimplement algorithms, and translate implementations across programming languages. Yet the resulting repository remains *derived*: these transformations can obscure source-specific provenance evidence without removing the underlying provenance relation.

This creates a new threat we call **repository-level agentic code laundering**: producing a derived implementation that retains the tested capabilities of a codebase while reducing the similarity evidence used to distinguish derivation from independent work. [Figure 1a](#) illustrates the setting. It matters for plagiarism detection ([Aiken, 1994](#); [Prechelt et al., 2002](#)), license-compliance auditing ([Inoue & Roy, 2021](#)), and software forensics, where similarity provides evidence of reuse.

Existing clone benchmarks primarily test fragments and bounded edits ([Svajlenko et al., 2014](#); [Roy & Cordy, 2018](#)), and adversarial perturbations test model robustness within existing programs ([Yefet et al., 2020](#)). Repository transformation benchmarks evaluate whether an agent can produce a func-

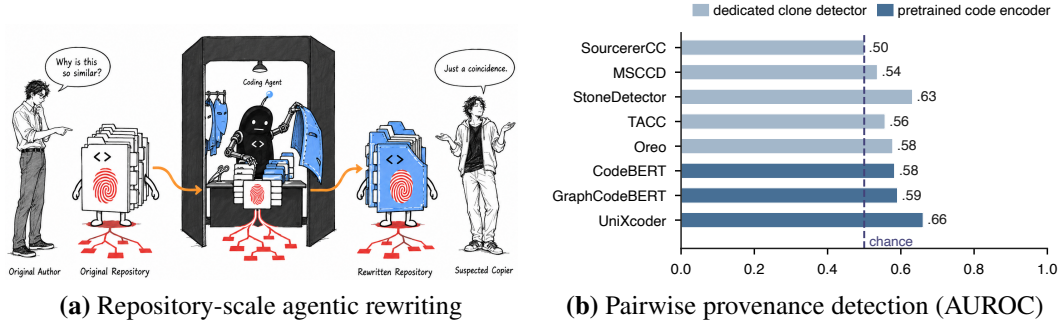


Figure 1: (a) Repository-level agentic code laundering: a coding agent transforms a source repository into a rewrite that preserves tested behavior while obscuring source-specific provenance evidence. (b) Existing methods on our pairwise provenance detection task (AUROC). **Light bars** denote dedicated clone detectors with supported-input coverage; these records are provisional and have incomplete coverage. StoneDetector, TACC, and Oreo support only Java→Java. **Dark bars** denote pretrained code encoders evaluated on the full benchmark. The dashed line indicates chance-level discrimination (AUROC = 0.5). The provenance relation remains even when detection is difficult.

tionally valid implementation (Wang et al., 2026; Li et al., 2026), but do not establish whether the resulting repository can still be traced to its source. Existing semantic-clone detectors degrade substantially under semantics-preserving transformations (Xu et al., 2026), suggesting an even greater challenge under repository-scale agentic reconstruction.

Nearest-source ranking alone does not answer that question either. Independently developed repositories in the same domain can share interfaces, vocabulary, and implementation patterns. A detector may rank the true source first without distinguishing it from such independent code. Provenance evaluation therefore needs behavior-verified rewrites, controlled non-source comparisons, and an evidence requirement that permits abstention.

We introduce Launder-Bench, a repository-level provenance benchmark with execution-verified rewrites across same-language and cross-language axes. Launder-Bench-Pair tests shared provenance between repositories; Launder-Bench-Attr tests source recovery among candidates with an evidence requirement. These are complementary views of the same provenance records. Measured same-domain independent repositories provide a source- and detector-specific reference for similarity without known derivation. This design separates relative retrieval from evidence that exceeds the independent-code reference.

Our contributions are as follows:

- *Problem formulation* (Section 2.1). We formulate repository-level agentic code laundering as a provenance problem: derivation persists even when rewriting obscures the evidence available to a detector.
- *Benchmark* (Section 2.2). We construct complementary Pair and Attribution tasks from behavior-verified agentic rewrites, with measured same-domain references and fixed candidate sets.
- *Empirical findings* (Section 3). The strongest full-coverage static detector reaches AUROC 0.661 and best F1 0.394. Encoders retain up to 22% Rank@1 but only 1–2% calibrated attribution, exposing a gap between relative similarity and source-specific evidence.

2 LAUNDER-BENCH

Launder-Bench evaluates provenance after coding agents transform real-world repositories while preserving tested behavior. Pairwise detection and source attribution provide complementary views of this problem.

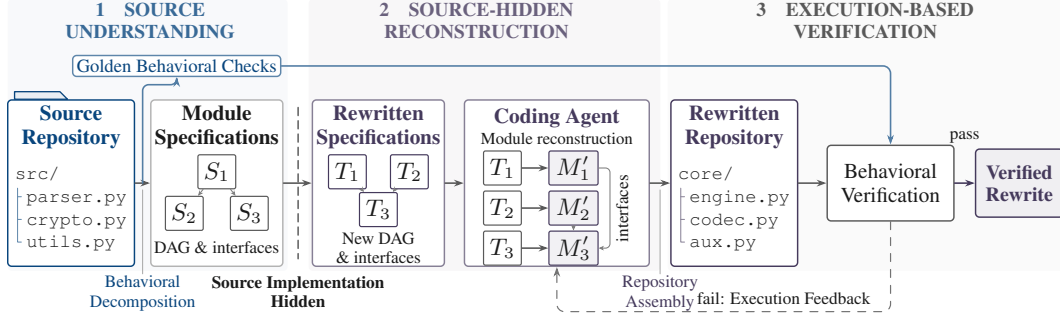


Figure 2: Agentic repository transformation. The source is decomposed into behavioral module specifications and executable checks, then hidden; a coding agent reconstructs the repository from restructured specifications and repairs it under execution feedback. Only rewrites that pass the behavioral checks are retained. DAGs and file names are schematic.

2.1 TASK FORMULATION

Let π denote an agentic rewriting policy and $\mathcal{P}_q$ the provenance set of source repositories provided to it. The policy produces a rewritten repository $R_q \sim \pi(\cdot \mid \mathcal{P}_q)$ whose tested functionality is preserved. The released benchmark instantiates the single-source setting, where $|\mathcal{P}_q| = 1$.

Pairwise provenance detection. Let $\mathcal{P}(R)$ denote a repository’s source-ancestor set in the benchmark provenance records. For an original source R_s , $\mathcal{P}(R_s) = \{R_s\}$; for a rewrite R_q , $\mathcal{P}(R_q) = \mathcal{P}_q$. Given (R_q, R_c) , the task is to determine whether $\mathcal{P}(R_q) \cap \mathcal{P}(R_c) \neq \emptyset$. This includes direct source–rewrite pairs and rewrite–rewrite pairs with a shared source. A detector assigns a provenance score $d(R_q, R_c)$, where a larger score indicates stronger evidence of shared provenance. We evaluate pairwise discrimination using AUROC and best F1, with decision metrics detailed in [Appendix F.1](#).

Source attribution. Given R_q and a candidate corpus $\mathcal{C}$, the task is to recover the source in $\mathcal{P}_q \subseteq \mathcal{C}$ by ranking candidates according to $d(R_q, R_c)$. We evaluate source retrieval and whether its evidence exceeds the source-specific independent-code reference band, as detailed in [Appendix F.1](#).

2.2 TASK CONSTRUCTION

We construct both tasks from verified rewrites and label them by provenance paths rather than surface similarity. The design varies provenance, functionality, and domain relatedness so that a detector cannot succeed merely by recognizing the task or programming language.

Repository transformation. We define rewriting independently of any particular model or coding harness. The agentic repository transformation is illustrated in Figure 2: the source is decomposed top-down into behavioral specifications and reconstructed bottom-up into a verified repository. The agent first decomposes the source repository into behavioral module specifications, their dependency structure, and executable checks (Line 2). The source implementation is then hidden. The backend restructures the specifications into target modules with a new dependency DAG and interface contracts (Lines 4–5). Each target module is rewritten bottom-up from its specification and the interfaces of already reconstructed dependencies (Lines 6–9). The rewritten modules are assembled into a new repository (Line 11), which is verified against the behavioral checks and iteratively repaired from execution feedback until verification succeeds (Lines 12–16).

Algorithm 1 therefore fixes the information flow: source-derived decomposition, source-hidden reconstruction, and execution-verified repair. Implementation details, including the rewriting backend, bounded repair recipes, and frozen artifacts, are provided in Appendix A.2.

Algorithm 1: Agentic Repository Transformation.

Require: Source repository R_s , agent A
Ensure: Verified rewrite R_q

- 1: $\triangleright$ Decompose the source into behavioral module specifications.
- 2: $(\{S_i\}_{i=1}^m, E, G) \leftarrow A.DECOMPOSE(R_s)$
- 3: $\triangleright$ Hide the source, restructure specifications, and rewrite modules.
- 4: $HIDESOURCE(R_s)$
- 5: $(\{T_i\}_{i=1}^n, E') \leftarrow A.RESTRUCTURE(\{S_i\}_{i=1}^m, E)$
- 6: **for** $i \in \text{TOPOLOGICALORDER}(E')$ **do**
- 7: $I_i \leftarrow \{\text{INTERFACE}(M_j) : (j, i) \in E'\}$
- 8: $M_i \leftarrow A.REWRITE(T_i, I_i)$
- 9: **end for**
- 10: $\triangleright$ Assemble and verify the rewritten repository.
- 11: $R_q \leftarrow A.ASSEMBLE(M_1, \dots, M_n)$
- 12: $F \leftarrow \text{VERIFY}(R_q, G)$
- 13: **while** $F \neq \emptyset$ **do**
- 14: $R_q \leftarrow A.REPAIR(R_q, F)$
- 15: $F \leftarrow \text{VERIFY}(R_q, G)$
- 16: **end while**
- 17: **return** R_q

Launder-Bench-Pair. The pairwise design contains four relations. A *source-rewrite* pair is positive by direct derivation. A *rewrite-rewrite* pair is also positive because both implementations descend from the same source. The two negative relations separate distinct confounders. A *rewrite-independent* pair preserves purpose but removes the provenance path, making it the central hard negative. A *rewrite-unrelated* pair changes both purpose and provenance, providing an easier control for incidental similarity. Together, these relations test whether a detector identifies derivation rather than shared functionality, generator style, or repository category. The relation definitions are summarized in Appendix A.4.

Launder-Bench-Attr. Each verified rewrite is queried against a fixed candidate set containing its true source exactly once. Source-matched independent repositories create the hard alternatives, while additional distractors prevent the task from reducing to a single pairwise comparison. Candidate language is matched where possible so that language identity does not reveal the source. A detector must therefore rank the complete set and may abstain. Attribution is credited only when the source is retrieved and its score exceeds the source-specific independent-code reference band. Freezing the candidate sets and negative neighborhoods keeps the comparison identical across detectors. Complete composition, matching rules, and validity audits appear in Appendix A.4.

Both tasks use the same underlying provenance records. Pairwise detection tests a specified relationship; attribution tests recovery among competing repositories. Frozen task identifiers, candidate sets, and reference neighborhoods make these comparisons reproducible across detectors.

2.3 FEATURES OF LAUNDER-BENCH

Conventional clone and plagiarism benchmarks primarily evaluate similarity between localized code fragments under bounded transformations. In contrast, Launder-Bench studies whether provenance remains recoverable after coding agents reconstruct real repositories while preserving tested behavior. This setting gives the benchmark several distinctive properties (Appendix B.1).

Repository-scale provenance. The unit of evaluation is a repository rather than a function or code fragment. Provenance evidence may be redistributed across files, interfaces, dependencies, and

architecture, requiring detectors to aggregate weak evidence across a large codebase. The released sources span from small projects to repositories containing millions of code tokens (Section B.2).

Behavior-constrained agentic rewriting. Difficulty cannot be created simply by perturbing code until similarity disappears. Each released rewrite is reconstructed by a coding agent and must reproduce source-derived executable behavior before entering the benchmark. The guarantee is deliberately limited to tested behavior rather than full semantic equivalence, but prevents detector failure from being attributed merely to non-functional transformations.

Provenance-controlled hard negatives. Similarity alone is insufficient evidence of derivation. Launder-Bench compares rewrites not only with unrelated repositories but also with independent same-domain and same-purpose implementations. These controls preserve functionality, vocabulary, or repository category without preserving the source-provenance path, forcing detectors to distinguish derivation from generic semantic relatedness.

Evidence-aware attribution. Closed-set retrieval always has a nearest candidate, even when the true source is absent or indistinguishable from independent code. Launder-Bench therefore pairs ranking metrics with source-specific reference bands measured from independent repositories. This separates the ability to retrieve a relatively similar source from the stronger ability to establish source-specific provenance evidence and enables explicit abstention diagnostics.

Same- and cross-language provenance. The benchmark spans both same-language and cross-language rewriting axes. Same-language instances test reconstruction without translation, whereas cross-language instances additionally change language-specific lexical and syntactic cues. Together they test which provenance signals survive each setting.

3 EXPERIMENTS

Evaluated models. We evaluate a detector suite spanning four evidence families (Table 1): lexical and statistical repository similarities; token clone indexes (SourcererCC (Sajnani et al., 2016) and MSCCD (Zhu et al., 2022)); structural or learned clone detectors (the TreeCen-inspired AST surrogate (Hu et al., 2022), StoneDetector (Heinze et al., 2026), TACC (Wang et al., 2023), and Oreo (Saini et al., 2018)); and repository embeddings (CodeBERT (Feng et al., 2020), sequence-only GraphCodeBERT (Guo et al., 2021), and UniXcoder (Guo et al., 2022)). These families inspect different evidence and have different repository-level applicability limits. Implementation provenance, language scope, checkpoints, and repository adapters are detailed in Appendix C.3.1. Rewrites use Codex CLI 0.144.5 with GPT-5.6-Sol as the default generator backbone; additional backbone ablations use GLM-5.3 and DeepSeek-V4.1-Flash (Appendix D.6). The generator backend, the test procedure of each detector family, and our Java-specific adaptations are summarized in Appendix C.1.

Evaluation protocol. Every detector receives the same frozen query and candidate repositories and is evaluated on both tasks from Section 2.1: pairwise detection uses the 1,128 positive pairs and their same-domain negative pairs, whereas attribution uses the fixed 20-candidate sets for 229 queries and permits abstention. We report AUROC for pairwise discrimination and best F1 as a retrospective threshold summary, calibrated TPR/FPR under each source-specific independent-code band, and Rank@1 plus calibrated attribution for source identification; native region outputs are aggregated through fixed repository adapters, embedding models use frozen windowing and top- k pooling. Both calibrated decisions share one source-specific independent-code band, $\beta_d(s) = \max_{N \in \mathcal{N}_s} d(R_s, N)$, defined by the source’s measured same-domain neighborhood:

$$\underbrace{\hat{y}_d(q, c) = \mathbf{1}[d(R_q, R_c) > \beta_d(s)]}_{\text{pairwise decision}} \quad \underbrace{g_{\text{cal}}(q; d) = g_{\text{rank}}(q; d) \mathbf{1}[d(R_q, R_s) > \beta_d(s)]}_{\text{calibrated attribution}} \quad (1)$$

with g_{rank} the tie-aware Rank@1 credit of Appendix A.4. Calibrated attribution is source-conditional evaluation credit, not a measured deployment abstention rate. Source-absence decisions are evaluated separately as a diagnostic; provisional native rows lack validated reference bands. The complete input, adaptation, calibration, abstention, and execution-integrity contract appears in Appendix C.2.

Table 1: Complete static-detector suite. Common-protocol methods are evaluated on all 1,128 Pair instances and 229 Attribution queries. Methods marked $\dagger$ do not support valid outputs on the full benchmark and are therefore reported only as provisional native results; their source-specific reference bands are not validated, so calibrated metrics are omitted. Calibrated attribution requires the true source to be both top-ranked and above its reference band.

Family	Detector	Pairwise				Attribution	
		AUROC $\uparrow$	Best F1 $\uparrow$	Cal. TPR $\uparrow$	Cal. FPR $\downarrow$	Rank@1 $\uparrow$	Cal. attr. $\uparrow$
Lexical / statistical	Token TF-IDF Cosine	0.493	0.350	0.044	0.007	0.050	0.000
	Character 5-gram Similarity	0.412	0.297	0.009	0.028	0.007	0.001
	Repository Statistical Profile	0.490	0.301	0.014	0.032	0.000	0.000
	Token-set Jaccard	0.288	0.297	0.032	0.128	0.000	0.000
Token clone	SourcererCC † (Sajnani et al., 2016)	0.501	0.299	—	—	0.048	—
	MSCCD † (Zhu et al., 2022)	0.536	0.298	—	—	0.113	—
Structural / learned clone	AST surrogate	0.558	0.319	0.124	0.128	0.050	0.010
	StoneDetector † (Heinze et al., 2026)	0.632	0.464	—	—	0.062	—
	TACC † (Wang et al., 2023)	0.557	0.434	—	—	0.312	—
	Oreo † (Saini et al., 2018)	0.578	0.432	—	—	0.000	—
Repository embedding	CodeBERT (Feng et al., 2020)	0.583	0.318	0.074	0.034	0.060	0.010
	GraphCodeBERT (sequence-only) (Guo et al., 2021)	0.591	0.320	0.104	0.058	0.090	0.020
	UniXcoder (Guo et al., 2022)	0.661	0.394	0.023	0.004	0.220	0.010

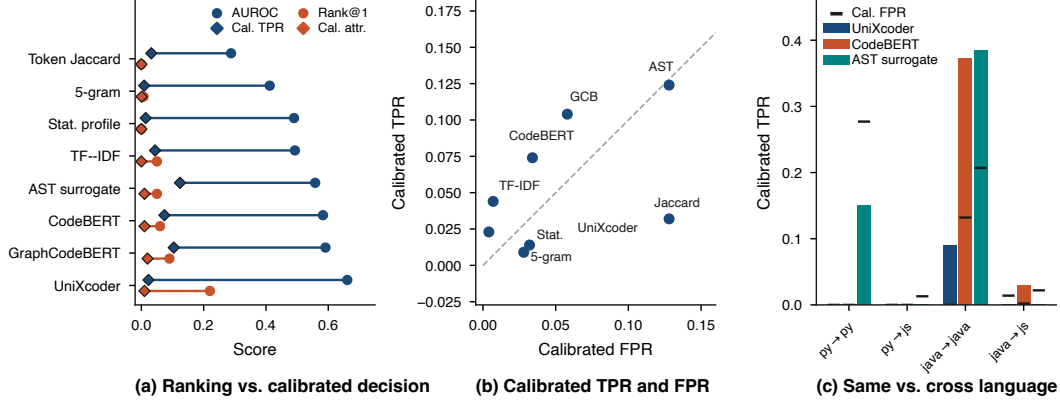


Figure 3: (a) Discrimination and retrieval alongside calibrated provenance decisions for the eight common-protocol detectors. Connectors pair AUROC with calibrated TPR and Rank@1 with calibrated attribution; AUROC and TPR measure different quantities, so their distance is not a performance drop on one metric. (b) Calibrated true- and false-positive rates per detector. The dashed line marks equal rates. (c) Calibrated TPR (bars) and FPR (ticks) on the four major language axes for three representative detectors. Same-language Java retains signal, with limited calibrated detection. Overall scores follow Table 1; Appendix C.3.2 gives the per-axis results.

3.1 MAIN RESULTS

Detector families retain different amounts of usable evidence. Table 1 separates detection performance from tool applicability. Lexical methods provide weak discrimination: no such method exceeds AUROC 0.493, and their calibrated attribution is at most 0.1%. Native clone tools have limited readable coverage and incomplete candidate sets. Their provisional AUROC ranges from 0.501 to 0.632, but readable coverage ranges from 60 to 368 Pair instances and from 11 to 69 Attr queries. Embeddings cover the full benchmark, with AUROC 0.583–0.661 and Rank@1 0.060–0.220, but calibrated attribution remains 0.010–0.020. Appendix F.1 maps each family to its evidence and applicability limits. The provisional native rows do not support a full-benchmark ranking against the encoders.

Ranking is not a provenance decision. UniXcoder ranks the true source first for 22% of queries, but only 1% receive calibrated attribution credit. GraphCodeBERT similarly reaches 9% Rank@1 and 2% calibrated attribution. Across the encoders, pairwise AUROC is 0.58–0.66, while calibrated

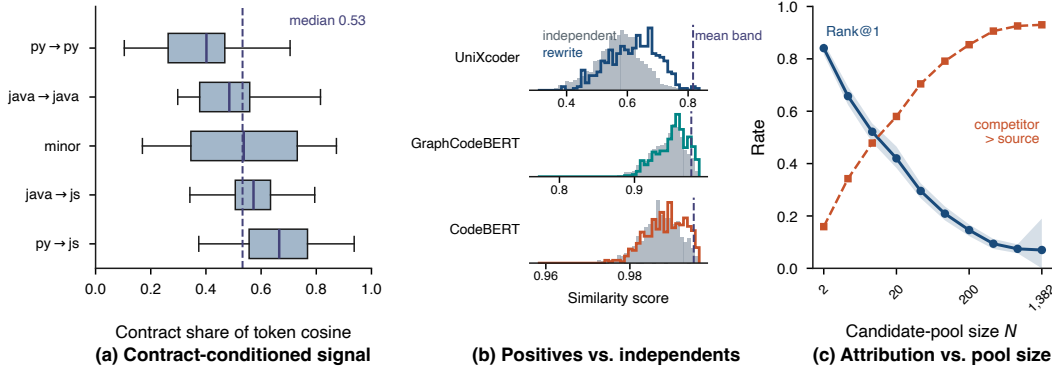


Figure 4: **(a)** Share of token-frequency cosine contributed by contract vocabulary, by language axis, with the remaining axes pooled. The dashed line marks the benchmark median 0.53 in the contract-vocabulary diagnostic; IDF weighting is omitted. **(b)** Rewrite-to-source and independent-candidate score distributions for the three embedding detectors, with the mean per-source band (dashed). The distributions overlap under each encoder; the mean band is a visual reference, not a pooled decision threshold. **(c)** Attribution Rank@1 and the rate of drawing at least one independent competitor scoring above the true source as the candidate pool grows toward the full 1,382-repository pool in the candidate-pool sensitivity analysis. The primary task fixes 20 candidates per query.

TPR is 2.3–10.4%. Thus relative source preference often fails the independent-code evidence requirement; ranking alone would miss this distinction. Figure 3a–b visualize the gap for every calibrated detector. See Table 1.

Translation amplifies the difficulty but is not required. The residual signal is strongest in same-language Java: UniXcoder reaches AUROC 0.834, but only 9.0% calibrated TPR. Detectors with higher TPR on this axis also incur FPR as high as 24.3%. More accepted positives therefore need not mean more source-specific evidence. The within-source comparison in Appendix D.4 also finds lower encoder margins after cross-language rewriting, while same-language margins remain low. Figure 3c compares calibrated detection across the four major axes.

Scale and input scope do not fully explain the gap. UniXcoder’s median margin remains negative across the source-complexity terciles. Restricting the source to oracle-anchored files changes its score by a median -0.033 , with few additional band passes. These diagnostics provide no broad recovery of calibrated detection through source-size stratification or removal of untested files (Appendix D.5).

Contract vocabulary explains part of the surviving lexical overlap. The token-frequency cosine diagnostic assigns a median share of 0.530 to tokens found in the behavioral contract. The share varies from 0.67 on Python-to-JavaScript to 0.40 on Python-to-Python, and is negatively associated with the UniXcoder margin on the four major axes ($\rho = -0.227$). This pattern is consistent with shared functional vocabulary preserving lexical overlap without reliably supporting a provenance decision. The diagnostic omits IDF weighting and does not decompose the primary TF-IDF score. Figure 4a shows the axis distribution. Appendix D.1.1 reports the axis breakdown.

Source absence exposes false attribution. Removing the true source does not force every detector to abstain. CodeBERT makes five false attributions and UniXcoder makes two across the 20 evaluated benchmark queries. A detector can rank candidates while lacking evidence that the source is present. Appendix C.4.1 separates correct, wrong, abstained, and false-attribution outcomes.

3.2 DISCUSSION

Repository provenance requires evidence beyond retrieval. The results distinguish the derivation relation from the evidence available in the rewritten artifact. A true source can remain the closest candidate without standing above independent same-domain code. The source-specific band

makes that distinction measurable, but does not establish a universal threshold or a deployable open-set decision rule. The source-absence study shows why retrieval must be paired with calibration and abstention ([Appendix C.4.1](#)).

Surviving similarity need not be source-specific. The detector families expose different limits. Lexical methods depend on overlap that reconstruction often reduces; native clone tools additionally require valid parsing, region mapping, and repository aggregation. Encoders recover more relative signal, yet shared contracts and domain conventions can also make independent implementations similar. The contract-vocabulary and candidate-pool diagnostics help explain why improving ranking alone may leave the provenance decision unresolved ([Figure 4](#)). These results motivate source-specific invariants alongside stronger representations, with calibration testing whether their signal also occurs in independent code.

Source-hidden reconstruction constrains behavior more than implementation. Execution fixes the tested inputs and outputs while leaving many internal choices open. The descriptor study records low source-to-artifact size and identifier overlap, alongside changes in file counts and structure ([Figure 5a](#)). Across the fully conforming rung cohorts, embedding band passage is already rare and lexical band passage is lower at the critic rung ([Figure 5b](#)). This pattern supports an information-bottleneck intuition: source-hidden reconstruction can retain behavior while discarding implementation-contingent details. It is neither an information-theoretic result nor evidence that every rewrite changes every structural feature; the rung cohorts also differ in membership.

Concrete behavioral contracts matter for reconstruction. In the 29-source generation study, purpose-only descriptions yield no rewrite passing the 0.99 conformance gate, whereas the contract-only arm reaches full conformance on 24/29 sources. The comparison concerns these specific recipes, but suggests that intended functionality alone underspecifies the tested behavior. [Appendix E.6](#) and [Appendix E.4](#) provide the designs and outcomes.

Behavior preservation limits simple surface edits. Across 20 artifacts, the tested identifier and string-literal edits all break behavioral checks. Comment laundering preserves behavior in 16/20 but changes TF-IDF and k -gram margins by at most 0.009. The tested interventions thus either break behavior or barely change the measured lexical evidence. [Appendix E.4](#) reports the complete intervention results.

The generator study bounds, rather than establishes, generality. The default GPT-5.6-Sol backbone and two alternative backbones produce valid rewrites that frequently fall below the seven evaluated static channels ([Figure 5c](#)). The pattern is therefore not confined to one generator in this study. The static suite’s failures do not imply that no detector can recover provenance, and the experiment does not isolate model capability as a causal factor.

The benchmark supports bounded provenance claims. Behavioral validity concerns the tested interfaces and inputs, and the same-domain references are screened rather than legally certified as independent. Native clone records remain provisional, and the primary calibrated-attribution metric is source-conditional credit. These boundaries motivate stronger hidden behavioral tests and validated repository adapters, with source-held-out calibration and explicit source-absence evaluation. The unseen-input audit ([Appendix B.3](#)) and detector protocol ([Appendix C.3.1](#)) specify what is currently supported.

3.3 CASE STUDY

Same-language Java remains difficult. The Java-to-Java slice removes language translation as an explanation. Of its Pair instances, 71% fall below every construction-channel band. TF-IDF accepts 24%, while UniXcoder accepts 9%. The failure therefore persists even when source and artifact share a language. The per-axis records are reported in [Appendix C.3.2](#).

More accepted positives also produce more false evidence. On this slice, CodeBERT and GraphCodeBERT recover 37% and 50% of positives. They also accept 13.2% and 24.3% of the

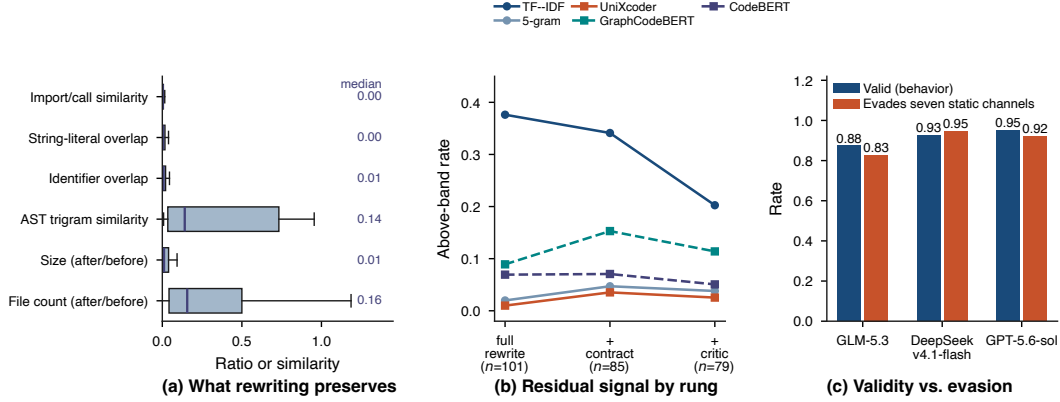


Figure 5: (a) Source-to-artifact descriptors across 1,156 recorded pairs: file-count and size ratios, AST-trigram cosine, and identifier, string-literal, and import/call similarities. AST values use 400 complete parses; medians are annotated. (b) Fraction of fully conforming rewrites above the source band, by rung of the rewriting pipeline ($n = 101, 85, 79$). These are separate accepted cohorts, not matched within-artifact changes. (c) Behavioral validity and provenance evasion among valid artifacts for three frontier generators under the same pipeline on 40 Python sources. Evasion refers to the seven static channels in the generator study, not the full primary suite.

same-domain negative pairs. The apparent gain is therefore not a clean recovery of source-specific provenance. The complete detector records are in [Appendix C.3.1](#).

Source removal exposes the decision error. Among the 15 Java attribution queries in the source-removal diagnostic, CodeBERT falsely names a remaining candidate five times. UniXcoder does so once. These errors show why ranking the available candidates is insufficient for provenance. [Appendix C.4.1](#) reports the complete case-study outcomes.

4 RELATED WORK

Code laundering. Detectors have long been stress-tested against transformations that hide reuse. Examples include obfuscation in software forensics and malware lineage ([Cesare et al., 2014](#); [Cheers, 2021](#)). Other work crafts semantics-preserving perturbations against neural code models ([Yefet et al., 2020](#); [Zhang et al., 2023](#)). Those evaluations use fragment-level benchmarks with mined or bounded synthetic edits ([Svajlenko et al., 2014](#); [Svajlenko & Roy, 2015](#); [2016](#); [Duncan et al., 2021](#); [Zhu et al., 2025](#); [Burock et al., 2025](#); [Wang et al., 2023](#)). Their ground truth is least reliable for semantic clones ([Krinke & Ragkhitwetsagul, 2022](#); [2025](#); [Roy & Cordy, 2018](#)). In Launder-Bench, the unit is the repository. Hard positives pass an execution criterion. [Appendix B.1](#) gives the detailed benchmark-level comparison and makes explicit which of these requirements are absent from prior evaluation settings.

Code plagiarism detection. Relevant methods include token fingerprints ([Aiken, 1994](#); [Prechelt et al., 2002](#); [Wang et al., 2018](#)), tree and graph matching ([Baxter et al., 1998](#); [Jiang et al., 2007](#); [Krinke, 2001](#); [Komondoor & Horwitz, 2001](#); [Liu et al., 2006](#)), clone indexes ([Cordy & Roy, 2011](#); [Sajani et al., 2016](#); [Saini et al., 2018](#)), and pre-trained code encoders ([Feng et al., 2020](#); [Guo et al., 2021](#); [2022](#); [Wang et al., 2021](#); [Tao et al., 2022](#); [Yu et al., 2019](#); [Büch & Andrzejak, 2019](#)). Stylometric attribution supplies another source of evidence ([Frantzeskou et al., 2010](#); [Caliskan-Islam et al., 2015](#); [Abuhamad et al., 2018](#)). All depend on some measurable lexical, structural, or stylistic trace. Neural clone detectors already generalize poorly to unseen functionality at fragment scale ([Liu et al., 2021](#); [Pinku et al., 2024](#)). Our tasks ask whether surviving relative similarity supports a calibrated claim at repository scale. SourceTracker/HST ([Gurioli et al., 2026](#)) retrieves source snippets for verbatim or renamed fragments. It uses Recall@N and MRR. Those metrics do not test source-specific reference-band decisions against same-domain repositories. LC³ ([Lan et al., 2026](#)) extends semantic

clone detection to long and cross-language code but still formulates the problem as clone matching rather than repository-level source provenance.

Agentic coding for repository transformation. Coding agents have progressed from function-level generation (Chen et al., 2021; Li et al., 2022) to modifying real repositories (Jimenez et al., 2024; Yang et al., 2024). Recent benchmarks evaluate repository transformation itself: RepoTransBench (Wang et al., 2026) studies cross-language repository translation with executable tests, while RepoMod-Bench (Li et al., 2026) evaluates repository modernization with implementation-independent behavioral tests. These benchmarks ask whether an agent can produce a functionally valid transformed repository. Our question begins after transformation: whether provenance remains detectable after agentic rewriting preserves tested behavior.

5 CONCLUSION

Coding agents make repository-scale rewriting a practical provenance problem: an implementation can be extensively reconstructed while retaining tested behavior and a real derivation path to its source. We introduced Launder-Bench to make this setting measurable, combining execution-verified agentic rewrites with independent same-domain controls and complementary Pair and Attribution tasks. Across the evaluated detector families, stronger representations recover some relative similarity, yet this signal rarely becomes source-specific evidence sufficient for calibrated attribution. The resulting ranking–decision gap is the central finding of our study: identifying the closest repository is not the same as establishing where code came from. Launder-Bench therefore shifts repository provenance evaluation from similarity alone toward evidence-aware detection with calibration and abstention. As coding agents increasingly transform existing software rather than generate isolated snippets, reliable provenance will require methods that recover source-specific invariants surviving such reconstruction. We release the frozen artifacts and evaluation protocol to provide a reproducible foundation for developing and testing those methods.

AI USE STATEMENT

We used generative AI to polish the manuscript and check its clarity and consistency. Coding agents were the primary tools for collecting and generating benchmark artifacts, including repository analysis, source-hidden rewriting, and execution-guided repair. AI tools also assisted with implementing and debugging experimental code, preparing figures, and checking result interpretations. The agent configuration, generator backbones, and prompts are documented in [Appendix A.2](#), [Appendix D.6](#), and [Appendix A.5](#). Generated rewrites underwent the execution-based verification described in [Appendix A.3](#). The authors take responsibility for the final manuscript, experimental results, and released artifacts, including content produced with AI assistance.

ETHICS STATEMENT

This work studies agentic code laundering for scientific evaluation of repository provenance and to support the development of stronger detection methods. We acknowledge that rewriting methods could also be misused to conceal copying or evade provenance checks. Our benchmark retains source identities and derivation records, and documents source license metadata ([Appendix A.1](#) and [Appendix B.2](#)). We intend the benchmark and pipeline for research on provenance detection; low detector scores do not establish independent authorship or justify disregarding source licenses. The reported provenance relationships are technical records of derivation, not legal determinations of ownership or infringement.

REPRODUCIBILITY STATEMENT

We will release the benchmark data and the complete pipeline code for data collection, agentic rewriting, behavioral verification, task construction, and detector evaluation. [Section 2](#) defines the tasks and construction protocol. [Appendix A](#) documents source selection, rewriting configurations, prompts, and acceptance checks; [Appendix C.1](#) and [Appendix C.2](#) describe implementation details and repository-level detector adaptations. [Appendix F.1](#) specifies the metrics, calibration, and statistical units. These materials, together with the frozen task manifests and evaluation records, support reproduction of the reported comparisons.

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A BENCHMARK CONSTRUCTION

This part documents how the benchmark was constructed: [Section A.1](#) the sources and their provenance, [Section A.2](#) the frozen agentic rewriting protocol, [Section A.3](#) behavioral verification and acceptance, and [Section A.4](#) the deterministic construction of the two tasks. [Section A.5](#) reproduces the concrete stage prompts. Launder-Bench combines repository-level provenance tasks with agentic rewriting and execution-based acceptance. The comparison in [Table 5](#) concerns this combination, rather than priority over repository transformation benchmarks in general.

Release policy: the full verified set. The published benchmark is the full frozen collection of construction-verified instances. Every released instance passed the same construction gates, and no detector-informed or difficulty-based selection defines the release: the “challenge tier” referenced in later analyses is a descriptive stratum of the released instances (below every construction band with substantive size), not a separately selected release set. A random control tier is released alongside the main instances so that selection-conditioned claims can be checked against unselected pairs.

Functional verification of every rewrite. A rewrite enters the benchmark only after it is verified to preserve its source’s tested behavior to the stated acceptance threshold. Verification is execution-based, not model-judged. A language-agnostic conformance harness first captures golden input–output traces by executing the *original* repository on a case list covering its core behavior; the rewrite is then executed on the same cases through a thin adapter that is excluded from all scoring. The release gates require the source harness to yield at least five successful cases, and the rewrite to reproduce at least half of the expected outputs exactly with overall conformance of at least 0.5; agreement on rejected inputs alone is insufficient. Two further validity gates reject degenerate evasion: the adapter must stay thin (at most half of the artifact’s tokens), and artifact size is judged against $\min(\text{original tokens}, 11,000)$. This capped denominator limits degenerate small artifacts without requiring whole-source size retention. Of the scored candidates, 1,243 pass the behavioral oracle and 1,232 remain after the adapter check; [Section A.3](#) reports the acceptance audits, including exact-match rates, an unseen-input audit, and a watermark scan of the released artifacts.

A.1 SOURCE SELECTION AND PROVENANCE

Dataset units and provenance. The published benchmark contains 282 pinned source repositories: 199 Python and 83 Java. Every source contributes 4 Launder-Bench-Pair instances, so the pairwise task spans $282 \times 4 = 1,128$ verified pairs. Launder-Bench-Attr queries are anchored on the sources that carry a full set of five measured same-domain reference repositories, because each attribution candidate set embeds the true source among these neighbors as hard negatives; 229 of the 282 sources satisfy this condition. Each attribution query contains 20 fixed candidates and the true source exactly once. The benchmark also contains the measured same-domain reference neighborhoods needed to reproduce every source-specific band. These neighborhoods are empirical controls. They are not certificates of legal or historical independence.

Source repository roster. The released manifest pins every source repository’s URL, commit, and snapshot date. The roster below lists all 282 sources, grouped by dominant code language (see [Figure 10](#) for their composition and scale). Names in blue mark the 229 sources that carry five measured same-domain reference repositories and therefore anchor Launder-Bench-Attr attribution queries.

Python (199): 3ES, ACES, AirbrakesV2, AlexaPi, Annif, AutoGPT, ChatGLM-6B, ComfyUI, Deep-Live-Cam, DeepSeek-V3, Fooocus, GitPython, GreedyBear, LLaMA-Factory, MONAI, MSS, MetaGPT, NeMo, OpenHands, PaddleOCR, PlasmaPy, Real-Time-Voice-Cloning, TLOmodel, TagStudio, TransformerLens, VDRS-Solutions, accelerate, agenta, aiohttp, aiortc, alumentations, ansible, ansible-cartridge, app, arkouda, backend, certbot, chainlit, checkov, ckan, conan, core, cpython, crosstl, cupy, cython, dask, dask-cuda, dgl, diffusers, django, django-celery-beat, django-ninja, django-two-factor-auth, docling, dsp, dvc, face_recognition, faceswap, fapolicy-analyzer, fastapi, faster-whisper, feast, flask, flet, fractal-tasks-core, gorilla, gpt-engineer, gpt4free, gpt_academic, grok-1, hackingtool, hal-cgp, imod-python, impactet, instructor, ivadomed, jax, joinmarket-clientserver, jwcrypto, kedro, keras, klein, knowledge-repo, kornia, langflow, langgraph, litellm, llama, localstack, locust, loguru, luigi, manage.get.gov, manim, matchms-backup, mathesar, matplotlib, meson, metaflow, micropython-lib, microsoft-authentication-library-for-python, mlflow, modin, moto, moviepy, mypy, nautobot, netbox, networkx, nltk, numba, numpy, oauthenticator, odoo, okta-jwt-verifier-python, open-MaStR, open-interpreter, openlibrary, openllmetry, openwisp-users, optimum-benchmark, optuna, paf, pandas, paperless-ngx, phidata, pip, plone.restapi, prefect, private-gpt, prowler, prymer, pulp_rpm, pydantic, pylint, pymapdl, pyopenssl, pypdf, pytorch, pytorch-lightning, qiskit, quiz-backend, ragas, raster-vision, ray, real-intent, repo_standards_validator, requests, rich, robotframework, roms-tools, rq, rucio, sancus-compiler, scapy, scikit-learn, scipy, scrapy, screenshot-to-code, securedrop-client, sentence-transformers, sgkit, sherlock, simtools, sisyphus, sktime, sopel, speechbrain, sphinx, sqlalchemy, sqlglot, stable-diffusion-webui, stcal, streamlit, supervision, sympy, synapse, tardis, tech-blog-hub-site, text-generation-webui, textec,

Table 2: Codex CLI 0.144.5 configuration with the default GPT-5.6-Sol backbone. Budgets are configured limits, not measured costs.

Stage	Contract / operation	Target / budget
Full contract	Extracted behavior with statistical camouflage	JS, Python, Java; ≤ 3 attempts
External vocabulary	Contract vocabulary shared across implementations	JS, Python, Java; ≤ 3 attempts
Critic	Feed surviving source names back to the agent	≤ 2 rounds $\times \leq 2$ attempts
Size repair	Expand the implementation against its specification	JS only; ≤ 3 attempts

tfl-live-tracker, thefuck, transformers, trl, ultralytics, uxarray, viztracer, vllm, xcdat, xclim, xformers, yfinance, yolov5, youtube-dl, yt-dlp, zarr-python, zulip

Java (83): JSONassert, JSqlParser, JavaHamcrest, JsonPath, alluxio, antlr4, assertj, avro, calcite, cassandra, checkstyle, commonmark-java, commons-beanutils, commons-cli, commons-codec, commons-collections, commons-compress, commons-configuration, commons-csv, commons-dbutils, commons-digester, commons-geometry, commons-imaging, commons-io, commons-jexl, commons-jxpath, commons-lang, commons-math, commons-numbers, commons-pool, commons-rng, commons-statistics, commons-text, commons-validator, doris, dubbo, elasticsearch, fastjson2, flexmark-java, flink, google-java-format, graphql-java, grpc-java, gson, h2database, hadoop, jackson-core, jackson-databind, java-diff-utils, java-jwt, javaparser, jcommander, jgrapht, jjwt, jline3, joda-money, joda-time, json-schema-validator, jsoup, jts, junit5, kafka, minimal-json, mockito, msgpack-java, mvel, nacos, netty, opensearch, picocli, pinot, rocketmq, sentinel, shardingSphere, snakeyaml, sonarqube, spotbugs, spring-boot, spring-framework, threeten-extra, typetools, univocity-parsers, zxing

Released artifacts. We release the rewritten repositories: for every accepted artifact, the rewritten project, its excluded conformance adapter, the oracle traces, the candidate lists, and the detector inputs, together with the pinned source manifest and the per-instance task definitions of both tasks.

Licenses and the released pipeline. The sources are third-party projects with the archived license metadata summarized in Table 6. The benchmark records technical derivation; it does not determine legal ownership or license compliance. To make the transformation process auditable, we also release the complete data pipeline: the agentic rewriting backend and its prompts (Section A.2 and Section A.5), the oracle-generation and conformance harness, the verification and selection gates (Section A.3), and the freeze manifests, so the benchmark can be audited, regenerated, and extended under the same protocol.

A.2 AGENTIC REWRITING PROTOCOL

Table 2 summarizes the Codex backend. The agent reads repository context, extracts a behavioral contract, constructs a new project with the source hidden, materializes the files, and executes the candidate. Execution failures and leaked identifiers are returned to the agent for bounded repair. The same backend is used across all language axes. The benchmark freezes the resulting repositories, adapters, oracle traces, candidate lists, and detector inputs.

A.3 BEHAVIORAL VERIFICATION AND ACCEPTANCE

Behavioral audit. Behavioral validity is oracle-covered rather than whole-program equivalence. On generation-visible successful cases, 62% of Pair artifacts reproduce every observed output exactly. The bounded benchmark unseen input audit gives 60/63 exact matches among inputs on which the original succeeds. By major axis, the counts are 14/14, 13/15, 16/17, and 17/17. These checks bound the claim to the tested interfaces and inputs.

Observed safety behavior and watermark audit. The research-context prompts describe a provenance benchmark and do not ask the agent to remove a watermark. Every Pair artifact contains code rather than a recorded policy refusal. We searched the benchmark artifacts, excluding code dependencies and oracle files, for case-insensitive explicit signatures including AI-generated, generated by OpenAI/ChatGPT/GPT/Codex, provenance watermark, and watermarked by; the audit returns zero matches. This is an output-level ob-

servation, not a claim about unlogged failed calls, imperceptible watermarks, or other providers. Its security implication is narrow but direct: the evaluated coding-agent workflow neither refused the observed rewriting requests nor volunteered a visible provenance signal on the scored outputs.

A.4 PAIR AND ATTRIBUTION CONSTRUCTION

Table 4 summarizes the composition of the two tasks.

Provenance paths, not surface similarity, define the labels. Let R be a source repository. In the positive branch, an agent first derives a behavioral specification D_R from R and then produces a rewrite R' :

$$R \longrightarrow D_R \longrightarrow R'. \quad (2)$$

Hiding the source implementation after extraction does not make R' independent: D_R is a source-derived representation, so the provenance path remains. Two separately generated rewrites R'_1 and R'_2 are likewise a positive pair because both descend from R .

The hard-negative branch requires a genuinely source-independent description $D_\perp$, such as a public task description or manually normalized functional requirements prepared without access to R or any representation extracted from it. Generation then follows

$$D_\perp \longrightarrow I, \quad R \not\rightarrow D_\perp, \quad (3)$$

and (R', I) is negative despite matched purpose. By contrast, generating I from a sketch extracted from R would create the path $R \rightarrow D_R \rightarrow I$ and must be labeled positive. This distinction prevents source hiding from being mistaken for provenance independence.

Table 3: Four conceptual Launder-Bench-Pair relations. The hard negative preserves functional relatedness while removing the source-provenance path.

Relation	Pair	Label	Meaning
Source–rewrite	(R, R'_1)	+	direct provenance
Rewrite–rewrite	(R'_1, R'_2)	+	shared provenance through R
Rewrite–independent	(R', I)	−	matched purpose without a path from R
Rewrite–unrelated	(R', N)	−	neither matched purpose nor provenance

The two negative relations form a difficulty hierarchy. (R', N) is an easy negative because functionality and provenance both differ. (R', I) is the central hard negative because functionality is held similar while provenance changes. The controlled independent-generation setting holds the coding-agent backbone, agent harness, compute budget, target language, and requested functionality fixed. It varies only whether the agent receives source-derived information or a source-independent functional sketch. This construction tests whether a detector identifies provenance rather than generator style, language, repository category, or generic coding conventions. The reference bands use pinned, measured same-domain repositories discovered independently of the rewrite process.

Launder-Bench-Pair instantiation. Each released query is a verified rewrite R' paired positively with its source R and negatively with measured non-source repositories. Same-domain neighbors provide hard negatives, while unrelated repositories provide an easier control. The neighbors are screened for copies and pinned at fixed commits, but their independence remains an empirical dataset assumption rather than a legal or historical certificate. The fixed release contains 1,128 positive queries from 282 sources over four rewriting axes. Every query retains the same source-specific negative neighborhood across detectors.

Launder-Bench-Attr instantiation. Each verified rewrite R_q defines an attribution query over a fixed set of 20 distinct candidates C_q containing its true source R_s exactly once. The remaining candidates are non-source repositories. Measured same-domain neighbors serve as hard negatives

Table 4: Launder-Bench composition. Pair and Attribution provide complementary evaluation views of the same provenance records.

Component	Size	Benchmark role
Source repositories	282	real projects; 199 Python, 83 Java
Same-domain negatives	1,244	measured; sim < 0.35; median 0.076
Launder-Bench-Pair	1,128	fixed repository-pair detection task
Launder-Bench-Attr	229	queries with 20 fixed candidates each; attribution with abstention

when available, and random distractors complete the set. Candidate identities and ordering are fixed across detectors, so every method solves the same retrieval problem.

Given detector score d , let

$$T_q(d) := \arg \max_{R_c \in \mathcal{C}_q} d(R_q, R_c) \quad (4)$$

denote the set of candidates tied for the highest score. Rank@1 gives the true source full credit when it is the unique maximum and fractional credit when it shares the maximum:

$$g_{\text{rank}}(q; d) = \frac{\mathbf{1}[R_s \in T_q(d)]}{|T_q(d)|}. \quad (5)$$

This is a closed-set retrieval measure: some candidate is always top-ranked, even when no score constitutes meaningful provenance evidence.

For each source R_s and detector d , the source’s measured same-domain neighborhood $\mathcal{N}_s$ defines the reference band

$$\beta_d(s) = \max_{N \in \mathcal{N}_s} d(R_s, N). \quad (6)$$

The true source passes the evidence gate only when its query score strictly exceeds this source-specific band, $d(R_q, R_s) > \beta_d(s)$. Calibrated attribution therefore requires both retrieval and evidence:

$$g_{\text{cal}}(q; d) = g_{\text{rank}}(q; d) \mathbf{1}[d(R_q, R_s) > \beta_d(s)]. \quad (7)$$

Thus Rank@1 measures whether the source is relatively closest among the candidates, whereas calibrated attribution asks the stronger question of whether that retrieval is also supported beyond the similarity already observed between the source and independent same-domain repositories.

Abstention and source absence. The evidence gate permits abstention when no candidate exceeds the calibrated threshold. To test this behavior outside the source-present setting, we remove R_s from selected candidate sets while retaining the remaining 19 candidates and the original threshold. Predicting any remaining candidate is a false attribution; predicting none is a correct rejection. This diagnostic is reported separately from the primary source-present attribution score.

Benchmark coverage. The benchmark spans four rewriting axes: py→js, py→py, java→java, and java→js. The benchmark artifact, source, candidate-list, oracle, and score identifiers are frozen so that every reported result maps to a public task instance.

A.5 PROMPTS

This section reproduces the stage prompts of the agentic rewriting backend verbatim. They are the concrete instantiation of the repository transformation in [Algorithm 1](#): every stage is a separate coding-agent session, and the session that implements a module never sees the original source or the other modules. One rewriting cycle runs three stages: *decompose* ([Figure 6](#)), *rewrite* ([Figure 7](#)), and *compose* ([Figure 8](#)). A decomposition session reads the current project and writes a unit manifest: a bottom-up decomposition into 3–8 core units, each with a responsibility, a behavioral specification, an interface contract, the exact data strings its behavior fixes, and its dependencies. Each unit is then implemented in isolation by a fresh session that receives only that unit’s entry and the interfaces of the units it depends on. A composition session assembles the units into a working project and

iterates in a test loop against the behavioral conformance oracle, which runs the source’s captured golden cases on the evaluation machine, until conformance is 1.0. The cycle then repeats on its own output, so the decomposition is re-derived each cycle rather than inherited. After the final cycle, a critic session (Figure 9) removes the source-chosen tokens that survived, gated on conformance staying at 1.0. Braced placeholders are filled per unit or per cycle; every prompt is otherwise sent to the agent exactly as printed. The headless-session driver appends one further instruction to the compose and critic stages: “Work autonomously and keep iterating – do not stop to ask.”

Decomposition prompt. The placeholder {target} is the source repository on the first cycle and the composed repository on every later cycle, which is what makes the decomposition re-derived rather than inherited.

```
Decompose the Python project in {target}/ into its core modules so that each can be
rewritten independently by someone who cannot see the rest of the project.

Write units.json: a JSON array of the core units, ordered BOTTOM-UP so each unit depends
only on
units that appear before it. Each entry:

{"name": "<short lowercase slug, used as the module name>",
 "layer": <int, 1 = depends on nothing internal>,
 "responsibility": "<one sentence>",
 "behavior": "<precise observable behavior: inputs, outputs, error categories, defaults,
    boundary
    and edge cases, data formats, ordering>",
 "interface": "<exactly what this unit exposes to the units above it: function and class
    names,
    signatures, and the meaning of every argument and return value>",
 "data_strings": [<exact literal values this unit's behavior fixes>],
 "depends_on": [<names of units this one imports>"]}

Rules:
- Each unit must be fully implementable from its own entry alone. Its rewriter is told
  nothing about the rest of the project except the interfaces of the units it depends
  on.
- 'interface' is the contract between units and the only channel of coordination between
  them.
- Together the units must cover every behavior the conformance cases exercise.
- Use 3-8 units. Fewer for a small project.
- Do NOT record file names, directory layout, or the original project's name anywhere.

Write only units.json.
```

Figure 6: Decomposition prompt. The session reads the current project and emits a bottom-up unit decomposition with explicit interface contracts; no file names, directory layout, or original project names may be recorded.

Isolated unit implementation prompt. One fresh session per unit, started in an empty directory containing nothing but this prompt; the placeholder fields are filled from the unit’s entry in the unit manifest.

```
Implement one module of a larger Python project, in this directory.

This is your entire specification. You cannot see the rest of the project.

UNIT NAME (this is the module name): {name}
RESPONSIBILITY: {responsibility}
BEHAVIOR: {behavior}
INTERFACE YOU MUST EXPOSE: {interface}
EXACT DATA STRINGS THAT MUST APPEAR IN OUTPUT: {data_strings}

Write this unit as a top-level Python module '{name}.py' (or a package directory '{name}/'
with
'__init__.py') in this directory, exposing exactly the interface above.
```

```

Dependencies: you may 'import' these sibling modules, which other people are writing in
parallel. Assume they exist at the project's top level and expose exactly these
interfaces: {dep_interfaces}

Choose your own internal names, data structures and algorithms -- nothing here constrains
them. Prefer the standard library; import a third-party package only if the behavior
genuinely requires it. Write real, runnable code, no TODOs and no placeholder stubs.

Do not create files other than your unit's.

```

Figure 7: Isolated unit implementation prompt. The session sees only its unit’s contract and the interfaces of the units it depends on, never the source or the sibling modules.

Composition prompt.

```

The units in units/ were rewritten independently, each by someone who could not see the
others. Assemble them into a working Python project in repo/.

- Copy every unit's files in, preserving their module names.
- Resolve interface mismatches by adjusting the CALLERS to match what each unit actually
exposes. Only rewrite a unit's internals if it plainly does not do what its contract
says.
- Add the glue the project needs: the conformance entry point 'repo/conf_cli.py', and
whatever wiring connects the layers.
- Build bottom-up: get the lowest layer importable and correct first, then each layer
above it.

The entry point must read the JSON case array on stdin and write the JSON result array on
stdout, and run as 'python conf_cli.py'.

TEST LOOP: run 'bash oracle.sh'. It copies repo/ to the evaluation machine and prints
conformance plus the concrete cases that disagree. Iterate until conformance is 1.0
and n_failing is 0.

```

Figure 8: Composition prompt. The session assembles the independently rewritten units and iterates against the behavioral conformance oracle until every golden case passes.

Critic prompt. Run once after the final cycle; `leaks.json` lists the source-chosen tokens that survived the rewrite.

```

repo/ still contains identifiers or strings that belong to ANOTHER project -- the one
whose behavior this project reproduces. They are that project's own naming choices,
not anything the behavior requires, and they must not appear in independently written
code.

The surviving tokens are listed in leaks.json. Remove every one of them: rename
identifiers to your own unrelated terms everywhere they occur, and reword any comment
, message or documentation string containing them. Do NOT change observable behavior,
and do NOT touch values the specification marks as externally required (protocol
fields, third-party API keys, file-format field names). If one of these tokens
appears in a value the conformance cases compare against, KEEP it -- behavior wins.

TEST LOOP: run 'bash oracle.sh' after each substantial change. Conformance must stay 1.0.

```

Figure 9: Critic prompt. The literal list of surviving source-chosen tokens is fed back; a round is accepted only if the conformance oracle still passes every case.

B BENCHMARK CHARACTERIZATION

B.1 COMPARISON WITH EXISTING EVALUATIONS

Prior evaluations provide important fragment-level or transformation-level tests, but they do not combine a repository-scale, source-hidden agentic rewrite with a provenance task, an independent

Table 5: Launder-Bench against related code-transformation and provenance evaluations. “Source-hidden” indicates that reconstruction proceeds without access to the source implementation; “Independent control” uses related but provenance-independent implementations; “Abstention” allows a detector to reject insufficient provenance evidence rather than force an attribution.

Benchmark	Repo. scale	Agentic rewrite	Source-hidden	Provenance task	Independent control	Abstention
BigCloneBench (Svajlenko et al., 2014)	–	–	–	✓	–	–
Mutation banks (Duncan et al., 2021)	–	–	–	✓	–	–
Adv. perturbation (Yefet et al., 2020)	–	–	–	–	–	–
SourceTracker/HST (Gurioli et al., 2026)	–	–	–	✓	–	–
RepoTransBench (Wang et al., 2026)	✓	✓	–	–	–	–
RepoMod-Bench (Li et al., 2026)	✓	✓	–	–	–	–
Launder-Bench (ours)	✓	✓	✓	✓	✓	✓

same-domain control, and abstention as an allowed decision. Table 5 summarizes these differences; the entries describe evaluation design rather than the general capabilities of the underlying detectors.

What each prior evaluation measures.

- **BigCloneBench** (Svajlenko et al., 2014) is a curated benchmark of inter-project clone pairs mined from roughly 25,000 Java projects. Its unit is a single function, and pairs are tagged by clone type from Type-1 through Type-4. The Type-3 and Type-4 labels rest on human judgment over mined candidates, so the ground truth is least reliable exactly for the semantic clones that a laundering benchmark must contain (Krinke & Ragkhitwetsagul, 2022; 2025).
- **Mutation banks** (Duncan et al., 2021) generate clones by applying a fixed catalog of syntactic mutation operators to a seed fragment. The transformation space is bounded by the operator list, so a detector can succeed by recognizing the operators rather than by finding provenance evidence. Positives carry no behavioral check beyond the operator’s construction.
- **Adversarial perturbation** (Yefet et al., 2020) crafts gradient-guided edits, such as identifier renamings, that flip a neural code model’s prediction on a snippet. This setting measures model robustness rather than provenance: the transformation is minimal by design, the victim model and ground truth are fixed in advance, and there is no notion of an independent reimplementations against which a detector should abstain.
- **SourceTracker/HST** (Gurioli et al., 2026) retrieves source snippets for verbatim or renamed fragments and is scored by Recall@N and MRR. The unit is again a fragment, and these ranking metrics do not test source-specific decisions against same-domain independent repositories.
- **Repository transformation benchmarks** (Wang et al., 2026; Li et al., 2026) evaluate agentic cross-language translation and repository modernization against executable or implementation-independent behavioral tests. Both operate at repository scale with agentic rewriting, but the agent reads the source during the transformation, and the measured quantity is the functional validity of the result rather than its provenance.

Why Launder-Bench differs. Three requirements jointly separate Launder-Bench from these settings. First, the unit is the repository and the transformation is an agentic rewrite that proceeds without access to the source implementation: the artifact is a source-hidden reimplementations under a tested contract, not a mined pair, a catalog mutation, a gradient perturbation, or a translation that reads the source. Second, the task is provenance rather than functional validity, and every positive must pass an execution-based conformance oracle on source-derived cases. Third, calibrated detection and attribution use measured same-domain independent references and permit abstention: ranking the true source first is not sufficient for calibrated credit, and a detector may reject insufficient provenance evidence. Native rows without validated bands remain explicitly uncalibrated. The evaluations in Table 5 do not combine all these requirements in a repository provenance task.

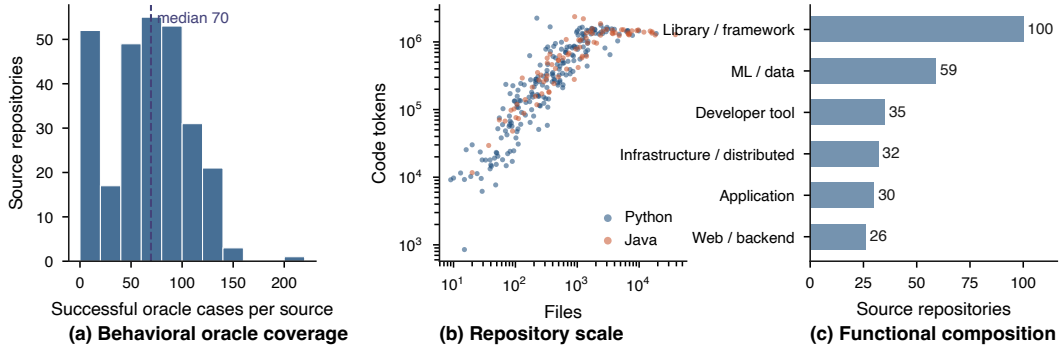


Figure 10: Source repository evidence, scale, and composition over the 282 released sources. (a) Number of successful behavioral oracle cases available per source repository, summed over its verified artifacts (median 70; 43 sources have no verified artifact). (b) Scale of the original repositories: files versus code tokens (log-log), spanning 851 to 2.4×10^6 tokens (median 3.5×10^5) and 9 to 38,532 files (median 412). (c) Functional composition of source repositories across libraries, applications, developer tools, ML/data systems, and infrastructure.

B.2 SOURCE REPOSITORY STATISTICS

Table 6 summarizes archived GitHub license metadata for the 1,510 repositories represented in the benchmark: the 282 sources and 1,228 further repositories from the negative list. The metadata snapshots may postdate the pinned revisions, and an SPDX identifier records a declared license, not a check of every file.

Table 6: Declared licenses from archived repository metadata. “GPL family” combines GPL, AGPL and LGPL identifiers; “other SPDX” combines MPL, EPL, Unlicense, Creative Commons, 0BSD, WTFPL and MIT-0.

License metadata	Sources	All required
Apache-2.0	120	389
MIT	50	410
BSD-3-Clause / BSD-2-Clause	21	51
GPL family	27	131
Other SPDX	5	25
SPDX identifier, total	223	1,006
NOASSERTION	36	127
No license field or metadata	23	377
Repositories	282	1,510

Composition, scale, and behavioral evidence. **Figure 10** summarizes the source pool. Every source repository is pinned to a specific commit in the released manifest, and all construction and evaluation ran against those pinned revisions. The sources are real, active projects and are large: originals span roughly three orders of magnitude in size, with a median of 3.5×10^5 code tokens across 412 files. Repository scale is therefore itself a detection variable, which **Section D.5** examines. Behavioral evidence backs the pool at scale: verified artifacts carry 18,381 successful golden oracle cases in total, a median of 70 per source, and every verified artifact clears the 5-case acceptance gate of **Section A.3**. Functionally, the pool is not concentrated in any single domain: libraries and frameworks (100), ML and data systems (59), developer tools (35), infrastructure and distributed systems (32), applications (30), and web and backend projects (26), with the largest category covering 35% of sources.

Table 7: Bounded unseen-input audit on 12 Pair instances. Only inputs that succeed on the original count toward agreement. Exact uses serialized JSON; numeric equivalence additionally treats values such as 0 and 0.0 as equal.

Axis	Sources	Inputs	Orig. success	Exact	Numeric equiv.
Python→Python	3	18	14	14	14
Python→JS	3	18	15	13	13
Java→Java	3	18	17	16	16
Java→JS	3	18	17	17	17

B.3 FUNCTIONAL PRESERVATION

We audit 12 Pair instances from ten sources, with three instances from each major language axis. Six new inputs were authored for each contract interface, and none duplicates a visible conformance input. Original and rewritten code ran in isolated copies. The inputs were authored by this project’s working agent, which is a limitation rather than a claim of independent testing. Of the 72 inputs, 63 succeed on the original and 60 of these match exactly; numeric equivalence adds none. The three mismatches are a dask Python→JavaScript rewrite that rejects a valid `evaluate_graph` input and returns a dependency list in reversed order, and a commons-jexl Java→Java rewrite that tags an integer result as `bigint` rather than `long`. This small deterministic sample tests unseen inputs within generator-visible contract slices, not new functionality or whole-project behavioral equivalence.

Static inspection of the 12 excluded entrypoints records their hashes and named calls into included implementation modules. Nine handle stream/JSON I/O and dispatch; AutoGPT, ComfyUI and commonmark-java also map operation or field names. AutoGPT inserts argument defaults, and ComfyUI recursively substitutes output strings and normalizes exceptions. Thus the adapters are not uniformly pure byte transport: presentation and boundary behavior can be implemented in excluded code. The audit identifies these responsibilities and delegation boundaries; it does not certify all internal behavior or justify treating an adapter token-ratio gate as a semantic proof.

B.4 REWRITE MAGNITUDE AND STRUCTURAL CHANGE

Table 8 complements the matched comparison in Table 10. It retains each detector’s native score scale: statistical-profile scores can be negative, so they should not be placed on the same radial scale as cosine or Jaccard similarities. For each rewrite, we first take the median score across its measured same-domain independent candidates; both displayed groups therefore contain one observation per rewrite. Across-rewrite summaries can conceal within-query ordering, which is why the teaser instead averages the fraction of independent candidates outranked by the true source. Neither view treats these same-domain controls as random repository pairs.

Table 8: Definition and interpretation of the native-score comparison over all 1,128 benchmark rewrites. This preserves the evidential content of the raw-score diagnostic without comparing incompatible detector scales.

Element	Definition
Rewrite-source group	One native detector score for every benchmark rewrite
Independent group	Within-rewrite median over measured same-domain candidates, then one observation per rewrite
Summary	Across-rewrite median and interquartile range; intervals are descriptive, not confidence intervals
Scale	Native per-detector scale; statistical-profile scores may be negative and are not pooled with cosine or Jaccard
Scope	Five repository-wide channels used in the matched diagnostic, not the complete clone-detector suite

Table 9: Information-access controls on 29 sources. Δ compares source-visible and contract-only outputs on the 15 sources with executable source-visible outputs, with Wilcoxon signed-rank p and within-source AUROC. Band pass counts the 58 contract-only outputs above the source-specific band. Executability does not imply acceptance; contract-only does not imply provenance independence.

Detector	Δ median	Wilcoxon p	AUROC	Band pass
TF-IDF	+0.055	6e−5	1.00	0/58
k-gram	+0.008	1e−3	0.83	5/58
Token Jaccard	+0.002	9e−4	0.87	6/58
Stat. profile	+0.549	0.035	0.67	2/58
UniXcoder	+0.088	3e−4	0.93	1/58
CodeBERT	+0.001	0.083	0.55	12/58
GraphCodeBERT	+0.003	0.135	0.63	13/58

B.5 INFORMATION-ACCESS CONTROLS

Design. This diagnostic compares the information supplied to the rewriter on 29 sources. Two contract-only implementations (C_{io} , two seeds) receive a purpose description, conformance inputs, and golden outputs without source implementation access. The source-visible $B_{inplace}$ arm rewrites the implementation in place; B_{spec} reuses a frozen specification-based rewrite. The latter is a reference arm, not a new matched-generation run. The contract-only arms receive source-derived outputs, so source hiding alone does not establish provenance independence under [Section A.4](#). We interpret them as information-access controls, separately from the screened same-domain reference repositories.

Execution success is distinct from acceptance. All 58 contract-only attempts execute, with median conformance 1.0. The source-visible arm executes on 15/29 sources, with median conformance 0.18 among those runs. These execution counts are not acceptance counts. The comparison concerns the tested recipes; it does not isolate source visibility as the cause of behavioral failure.

Information access changes the residual similarity. The source-visible arm has a positive median margin advantage over the contract-only arms on every channel ([Table 9](#)). Separation is strongest for TF-IDF (AUROC 1.00) and UniXcoder (0.93); CodeBERT and GraphCodeBERT show weaker differences with $p > 0.05$. These comparisons use the 15 executable source-visible outputs, rather than 29 behavior-verified positives. Band passage among the 58 contract-only outputs is low for the lexical channels and reaches 13/58 for GraphCodeBERT. Low band passage does not establish independent origin.

The frozen specification-regeneration arm has a negative median advantage relative to the contract-only controls on every channel (UniXcoder -0.025 , $p = 0.027$). The two contract-only implementations also resemble each other (median TF-IDF similarity 0.28) more than a source-visible rewrite resembles its source (0.17). These observations support the importance of supplied behavioral information, while the different behavioral validity and generation histories limit causal interpretation.

C DETECTOR EVALUATION

C.1 IMPLEMENTATION DETAILS

Generator. Every release rewrite is produced by the Codex CLI 0.144.5 coding agent with GPT-5.6-Sol as its default generator backbone, driven through the staged contract, external-vocabulary and critic recipes of [Appendix A.2](#). The generator study of [Section D.6](#) re-executes the identical pipeline with GPT-5.6-Sol and two alternative backbones, GLM-5.3 and DeepSeek-V4.1-Flash, while holding the source set and evaluation protocol fixed.

How each detector family is tested. The lexical and statistical channels read the staged repository text directly. The native clone tools inspect staged query and candidate repositories; SourcererCC and MSCCD use recorded multi-document batches, while the pairwise wrappers stage repositories side by side. Reported region pairs are filtered to cross-repository matches and aggregated into the token-coverage score of [Section C.2.1](#). The embedding detectors encode a frozen windowed view of each repository and pool the top query-window cosines ([Appendix C.2](#)). Every detector sees the same frozen query and candidate repositories; unreadable inputs carry explicit dispositions rather than zeros.

Java adaptations. StoneDetector, TACC and Oreo are Java-only, method-level tools, and we adapted them rather than scoring them out of scope. Eligible Java files are staged before any candidate comparison, so directory names cannot shrink the coverage denominator. TACC assigns function IDs through an unsynchronized static counter, so we run one native thread per comparison and parallelize across isolated JVMs. MSCCD builds token frequencies within each recorded batch, so its batch plans are retained as part of the operating point. These three tools are reported only on their readable Java→Java inputs ([Appendix C.3.1](#)).

C.2 BASELINES AND REPOSITORY-LEVEL ADAPTATION

Protocol boundary. The primary common-protocol results use the frozen score and band rules in [Section 3](#). Native clone records remain provisional. This section distinguishes implemented input and aggregation rules from a proposed source-held-out calibration contract. The proposed 1% FPR operating point and unique-top/unique-acceptance rules below do not define the primary calibrated-attribution column.

Original method, adapter and task. We distinguish a method’s original output from the repository adapter and from the final decision rule. Clone detectors return accepted region pairs; encoders return representations; a classifier may return only a label. Region coverage is our adapter, not a probability or a score supplied by the original detector. Encoder pooling is likewise a benchmark adaptation. The comparison includes provisional repository adaptations of clone detectors whose native outputs provide no repository ranking. We report native detection and adapted attribution separately; neither table is presented as a reproduction of the other task. We record tool/checkpoint revisions, native thresholds, supported languages, minimum unit sizes, preprocessing and resource limits. An AST-type-count cosine surrogate cannot be reported as TreeCen’s graph-centrality features and trained classifier. UniXcoder requires its documented encoder-only format; a GraphCodeBERT checkpoint without data-flow inputs is a sequence-only ablation. Such substitutions must be labelled separately.

Inputs and applicability. Methods draw from the same frozen query and candidate repositories and candidate sets, recorded with content hashes. We separately record each preprocessor’s effective file view, harness exclusions, parsing coverage and native unit filters; shared repositories do not imply identical inspected code. A Java-source-only encoder ablation isolates the effect of including documentation and configuration files in the repository view. The proposed source-held-out contract groups rewrites of one source and known forks or duplicates, and requires disjoint tuning, calibration, and evaluation source groups. Candidate-library reuse must also be audited. These requirements do not certify that the primary source-conditional bands implement a deployable source-held-out predictor. The existing random control tier is not assumed to be an independent calibration set. Matched sensitivity comparisons use the same valid queries within each language axis. The primary native rows instead report their own readable input populations, which need not coincide. Cross-language stress tests are separate from a method’s supported native scope. Unsupported input, no eligible units, parse/mapping failure, timeout and execution failure are distinct from valid execution with no match. They are never converted to numerical zeros. Attribution requires 20 distinct, valid candidate comparisons, with the source occurring exactly once. We also report failures against the intended population fixed before execution, with excluded IDs and reasons: the intersection of successful runs alone can conceal failures on difficult inputs. Resource caps, truncation, sampling and inspected-input fractions are recorded per method. At this audit stage, all 1,382 referenced repositories agree with the canonical host under the evaluated repository-text view. A separate na-

tive raw-byte audit verifies all 2,526 input views after transport metadata removal, including files beyond the text-view size cap. Neither audit establishes absence of semantic duplicates.

Region-output adapter. The adapter proposed for validation is directional query coverage. Let $U_d(q)$ be query token positions eligible under the declared input and unit rules, fixed independently of the candidate, and let $M_d(q, c) \subseteq U_d(q)$ be their union over valid cross-repository matches. Define

$$s_d(q, c) = \frac{|M_d(q, c)|}{|U_d(q)|}. \quad (8)$$

An empty denominator is unavailable. Overlapping ranges count once; file mapping and each tool’s line-coordinate convention must be checked. The denominator does not depend on candidate length, so adding unmatched candidate code cannot dilute existing query evidence. We retain native match/no-match outputs and report eligible-input fractions. Weighted whole-repository coverage and symmetric coverage remain named sensitivity analyses, not choices selected for favourable evaluation results. In the implemented staged-Java sensitivity, file eligibility is fixed before candidate comparison: all staged Java files up to 2,000,000 bytes. The scorer does not apply another directory-name filter after native staging. Thus a Java package named `env` contributes to the denominator regardless of whether a candidate matches it. Multiline lexical tokens that intersect several disjoint reported regions still count only once. These checks repair aggregation; they do not validate the proposed native-unit eligibility denominator. For embedding adapters, model-specific input recipes precede a frozen windowing and pooling rule. The diagnostic configuration uses 64 deterministically stride-selected windows, a 512-token limit and the mean of the top eight query-window nearest-candidate cosines. All such settings belong in representation-cache keys together with the checkpoint revision.

Execution integrity. Process success alone is insufficient: uncaught native worker failures can leave a tokenizer waiting indefinitely or produce incomplete output. We reject such output, retain errors and distinguish no eligible units from a valid no-match result. For MSCCD, bounded batches retain every requested repository pair; oversized pairs are not truncated. Native input/output maps, configuration hashes and successful shard caches support replay. Runtime changes to logging and queue polling require native-output equivalence checks on positive and negative fixtures. Fixture agreement is reported as a bounded regression check, not a universal equivalence claim. MSCCD constructs token frequencies within each batch, so batch membership is part of its operating point. We retain batch plans and do not infer partition invariance merely from retaining every requested repository pair. The unchanged TACC implementation assigns function IDs using an unsynchronized static counter. A constructor-level concurrency probe reproduced duplicate IDs. We therefore use one native thread per comparison and parallelize isolated JVMs.

Calibration without query labels. For each method and sufficiently populated language axis, the proposed operating point targets an empirical false-positive rate of at most 1% on calibration negatives. Require at least 100 negative pairs from 20 source groups; otherwise mark the axis uncalibrated. With the strict rule $s > \tau$, choose the smallest observed-negative threshold satisfying this constraint. This empirical constraint is not a population guarantee; report realised counts and source-cluster uncertainty. Hyperparameter selection uses tuning groups, while calibration sets only the threshold. No evaluation label, true-source identity or test-optimal F1 threshold may enter the prediction rule. The native/default operating point is reported separately.

Attribution, ambiguity and abstention. To align scored repository adapters, the proposed primary decision returns the uniquely highest-scoring candidate only if it exceeds its frozen applicable threshold. Otherwise it abstains. We distinguish no accepted evidence, tied highest scores and a highest-scoring candidate below its threshold. We do not substitute a lower-scoring candidate because it passes a different threshold. This requires validated comparability of candidate scores within a query; different methods need not share a numerical scale. Coverage scores are not provenance probabilities. This decision rule remains subject to source-held-out validation.

As a separately named conservative sensitivity analysis, for a query’s fixed candidate set, form

$$A_d(q) = \{c : s_d(q, c) > \tau_{d, \text{axis}(q, c)}\}. \quad (9)$$

Return the candidate only when $|A_d(q)| = 1$. Otherwise abstain, distinguishing no evidence ($|A_d(q)| = 0$) from ambiguity ($|A_d(q)| > 1$). This rule treats candidates symmetrically and does not inspect the correct answer. All-zero clone coverage gives abstention, not successful attribution. Report correct and wrong attribution, abstention by reason, answer coverage and accuracy conditional on answering; the last is unavailable at zero coverage. Forced Rank@1 is a separate diagnostic for scored adapters, accompanied by all-zero and top-tie rates. Fractional tie credit does not imply a successful identification. A binary-only native output is not assigned an invented continuous ranking. True-source rank times true-source band passage remains labelled source-conditional verification.

This conservative unique-acceptance sensitivity rule is an explicit benchmark adapter, not the original method’s attribution algorithm. It abstains even if one of several accepted candidates has a higher score. A unique-top or score-margin rule requires a prespecified and calibrated adapter; the choice cannot depend on challenge performance. Native hard-label methods retain their accepted candidate sets and are reported separately from scored coverage adaptations.

Pair-level FPR is not a query-level false-attribution guarantee. Even a population per-pair bound α would yield only the union bound 19α on accepting any of 19 negative candidates; an empirical calibration rate does not establish that population bound. We therefore measure wrong attribution and ambiguity directly on held-out candidate sets. Source-present tasks do not establish open-set rejection when the true source is absent. The proposed operating point and minimum calibration counts remain design choices under validation, not guarantees of statistical power or query-level error control.

C.2.1 SCORING CLONE-PAIR DETECTORS AT REPOSITORY SCALE

This section documents the repository-level coverage adapter. The protocol in [Appendix C.2](#) separates native evidence, eligible-query coverage, and calibrated prediction. Current SCC/MSCCD collection uses recorded multi-document batches; it must not be described as independent two-document runs.

The classical clone detectors of [Table 1](#) do not return a similarity between two repositories. Each returns a *list of clone pairs*: StoneDetector, TACC, Oreo and the AST surrogate emit method-level pairs $(f_a, s_a, e_a, f_b, s_b, e_b)$ of file and line ranges; MSCCD emits token-bag blocks with line ranges; SourcererCC in file-level mode emits whole-file pairs, treated as the range of the whole file. The protocol of [Section 3](#), however, needs one number per (query, candidate) pair for every metric it reports. The adaptation is a fixed aggregator from a pair list to a score, applied identically to every tool.

AST surrogate, not native TreeCen. The AST-surrogate implementation counts named tree-sitter AST node types per function and thresholds the cosine between count vectors at 0.85. It does not compute TreeCen’s graph-centrality features or execute its trained classifier. The archived upstream implementation instead constructs tree graphs and centrality features before supervised clone classification. Thus the AST-surrogate `treecen` scores, tables and neighbor sensitivity measure this surrogate only. The local parser helper returns an empty vector on exceptions and maps zero-norm comparisons to zero, without establishing a valid no-match result. Native TreeCen evaluation requires a pinned feature and classifier implementation, validated input/pair mappings and parsing, and self-copy and partial-copy controls before its scores enter comparison. The released audit materials record the implementation hashes and a read-only check of the retained score tables.

Cross-repository matches only. Pairwise wrappers stage a query A and candidate B side by side. SourcererCC and MSCCD instead use the recorded multi-document batch plans, retaining distinct project identifiers; they are not independent two-document runs. Its output is filtered to cross-repository pairs: a pair whose members both lie in A or both in B is a within-repository clone (accessors, boilerplate, fixtures) and says nothing about derivation. For native wrappers, filtering retains the reported cross-repository pairs; thresholds, minimum sizes and verification stages are the recorded settings (StoneDetector LCS 0.3, minimum function size 10; TACC $N=3$ with default seed and verify scores; Oreo’s released classifier; SourcererCC 70%; MSCCD 0.7 with 20 minimum tokens; AST-surrogate cosine 0.85). The surrogate threshold is a local choice, not an upstream TreeCen default.

Token coverage. Let $T(X)$ be the multiset of lexer tokens of repository X over the files the tool can read (each side lexed in its own language), keeping keywords, names, literals, operators and punctuation and dropping comments, whitespace and docstrings. For every file of A the line ranges of all surviving cross pairs are merged, and $M(A)$ counts the tokens of A inside a merged range; $M(B)$ likewise. The score is

$$\text{sim}(A, B) = \frac{M(A) + M(B)}{|T(A)| + |T(B)|},$$

the fraction of the two repositories' code tokens that the detector places inside some cross-repository clone. It is bounded in $[0, 1]$ and symmetric; the joint denominator prevents a small artifact from being rewarded for covering its own few files against a large candidate; whole-file, block and method pairs reduce to the same token scale, and a token is counted once however many pairs cover it, so a tool that reports many small pairs gains nothing. Only *which* code a tool claims is cloned enters the score, never a per-pair confidence: the object under test is the clone decision an auditor would act on.

The self-pair ceiling. On identical inputs a file-level tool covers every token (SourcererCC 1.000, MSCCD 0.9998 on a Java source against itself), whereas a method-level tool cannot: package and import lines, field declarations, class headers and the methods it discards as too small lie outside every pair, so the same self-pair scores 0.48 (StoneDetector), 0.38 (TACC) or 0.73 (Oreo). This ceiling depends on the tool and its adapter, so raw coverage scores are not directly comparable across tools. Source-specific bands are used only where validated for the reported protocol; the clean-input native rows in Table 1 have no calibrated results. AUROC and best F1 remain within-detector summaries over readable inputs.

Alternatives not used. An earlier harness scored *function coverage*, $\frac{1}{2}(|\text{covered units of } A|/|\text{units of } A| + |\text{covered units of } B|/|\text{units of } B|)$ over methods extracted by the benchmark's own parser. It weights a three-line accessor and a three-hundred-line parser equally, depends on an extractor external to the tool, and is undefined for whole-file tools; token coverage replaced it in the pairlist harness. Aggregation does not validate the tool or the mapping from reported pairs to source ranges. No pair is weighted by the tool's similarity value and no per-tool threshold is tuned on the benchmark. Every document is staged without the conformance harness and build outputs. Effective file views and eligible units still differ across native tools, construction channels, and embedding adapters.

C.3 COMPLETE PAIR RESULTS

The benchmark's three construction channels (lexical `tfidf_cosine`, character-level `kgram_jaccard`, statistical `stat_profile`) and the token-overlap measure `token_jaccard` are deliberately simple, unpublished instruments: they are the channels the construction itself uses for difficulty scoring and band calibration, and they double as strong sanity baselines. This section reports their full results.

C.3.1 COMPLETE DETECTOR RECORD

This section collects every detector record on the release in two blocks. The upper block holds the eight band-calibrated records on all Pair and Attr instances: the three encoders of Table 1, the three construction channels, token Jaccard and an exploratory AST surrogate (the `treecen` record, labelled explicitly throughout the statistical appendices). The lower block holds the five native clone detectors as provisional records on their readable inputs: SourcererCC and MSCCD read 368 and 363 Pair positives and 69 and 64 Attr queries; the Java-only StoneDetector, TACC and Oreo read 60, 63 and 63 Java→Java positives and 16, 16 and 11 of that axis's Attr queries. Every unread instance carries an explicit disposition. Neither block is a validation of published detectors, and the provisional values are not comparable with the band-calibrated records or with one another.

Among the eight band-calibrated records, calibrated TPR on the 1,128 Pair positives ranges from 0.009 to 0.124, while pooled AUROC ranges from 0.288 to 0.661. Thus the benchmark degrades ranking as well as calibrated decisions; it does not support a claim that all ranking metrics remain

Table 10: Relative evidence versus calibrated decisions on all 1,128 Pair instances. True-source superiority averages, per rewrite, the fraction of measured same-domain candidates scored below the true source; ties receive half credit. Intervals use 4,000 source-cluster bootstrap resamples.

Result	Observation	Consequence
Relative ordering	TF-IDF and UniXcoder retain source preference; three other point estimates fall below the 0.5 no-preference reference	Some relative signal survives rewriting
Calibrated detection	Only 0.9–4.4% clear the detector-specific independent-code band	Relative preference rarely becomes actionable evidence
Rewrite validity	Names, modules, and sometimes language change while oracle-covered behavior remains	A new implementation is not evidence of independent origin

near ceiling. The three highest band-pass rates require their negative-pair context: the AST surrogate exceeds the band on 12.4% of positives but also on 12.8% of same-domain negative pairs, the sequence-only GraphCodeBERT on 10.4% against 5.8%, and CodeBERT on 7.4% against 3.4%. The AST surrogate thus has nearly equal pooled TPR and FPR; the two encoders have a positive TPR–FPR gap, although all three yield conjunctive attribution on only 1–2% of Attr instances. Among the band-calibrated records, UniXcoder retains the strongest attribution ranking signal (Rank@1 0.220), but conjunctive calibrated attribution is only 0.010.

The distinction between ranking and decisions remains informative even when neither is strong. TF-IDF ranks the source first on 5% of Attr instances, but none of those instances also clears its band. Token Jaccard crosses the source band on 6% of attribution queries, yet never ranks the source first; counting band passage as attribution would therefore give a misleading success rate. We recompute the final attribution column from the recorded top-rank/tie information and band decisions, requiring both. The small nonzero k-gram value is fractional credit from one above-band 13-way top tie, not an unambiguous source identification.

The axis-level records show that TF-IDF band passage is 19/78 on Java→Java and zero on the other axes, and the same axis supplies 7 of UniXcoder’s 10, 29 of CodeBERT’s 32 and 39 of GraphCodeBERT’s 45 accepted positives. These are not only pairs the construction channels already flag: 2 of UniXcoder’s 7, 19 of CodeBERT’s 29 and 24 of GraphCodeBERT’s 39 Java→Java acceptances fall among the 55 pairs of that axis below the band on every construction channel. The encoders thus add evidence where the channels see none, but on this axis CodeBERT and GraphCodeBERT also accept 13.2% and 24.3% of same-domain negative pairs. No single rewriting recipe drives the result: over the full-contract, external-vocabulary and critic recipes (148, 139 and 146 positives), accepted positives are 15/18/12 for GraphCodeBERT, 13/12/7 for CodeBERT, 2/5/3 for UniXcoder and 8/7/4 for TF-IDF, differing by at most six for every detector without a consistent ordering. [Appendix C.3.2](#) gives the upper-block records other than the sequence-only GraphCodeBERT by major axis ([Table 11](#)).

The five provisional native records were collected under the clean-input protocol without reference bands: their pooled AUROC spans 0.501 (SourcererCC) to 0.632 (StoneDetector), and forced Rank@1 spans 0.000 to 0.312 (TACC) over their readable attribution queries. These are the native entries of [Table 1](#), with coverage not comparable with its common-protocol rows: Oreo’s AUROC rests on 63 Java→Java positives, the encoders’ on 1,128 positives across four axes. Their band-calibrated columns are dashes rather than zeros because that clean-input protocol has no validated bands; the native-output mapping audits and the parsed-units sensitivity ([Appendix C.2](#)) govern their validity boundary. Deleting the most influential calibration neighbor raises both the AST surrogate’s TPR and FPR on eligible queries ([Appendix F.1.1](#)); the empirical band is sensitive to neighborhood composition.

[Table 10](#) makes the detection problem concrete. Across the 1,128 benchmark rewrites, k-gram, statistical-profile and token-Jaccard scores favor the true source over same-domain independent candidates less than half the time, with ties receiving half credit. TF-IDF and UniXcoder retain relative signal, but only 0.9–4.4% of rewrites exceed the source-specific independent-code band on

any one of these five channels. The finding is therefore not that every similarity signal disappears: relative similarity can survive without exceeding the empirical source-group threshold. This is a failure under that decision rule, not an absence of all possible provenance evidence. These values characterize the benchmark and are not a success rate for arbitrary rewriting attempts. [Section 3](#) reports the encoder comparison and the distinction between ranking and grouped-band decisions.

C.3.2 STRATIFIED AND AXIS-LEVEL RESULTS

All statistics in this section reuse the frozen per-instance records; no model, detector or behavioral test was rerun. Pair has 1,128 instances from 282 sources and Attr has 229 from 229. Pair quality tiers span hard, borderline, medium and stub: the single stub-tier instance (`java-jwt`, capped size ratio 0.187) enters through the Java→Java axis and has $d_{\text{pair}} > 0$, so it is not in the challenge tier. For each task, we draw 4,000 bootstrap samples of its sources with replacement, retaining all instances of each sampled source and recomputing the instance-weighted rate (NumPy PCG64, seed 20260916, sorted source names). Brackets give percentile 95% intervals conditional on the fixed artifacts, bands, and task manifests. They do not account for generation, neighbor selection, or unseen-source bias; zero-success intervals are degenerate bootstrap intervals, not population upper bounds. We use stored decisions and exact top-tie counts rather than rethresholding rounded scores. AUROC, best-F1, FPR and Rank@1 intervals are not estimated here. The AST surrogate is the `treecen` record, not the published TreeCen method; score completeness does not certify native-tool validity.

Strict-positive sensitivity. Restricting Pair to full positive-case conformance leaves 62% of instances. Of these, 45% meet the challenge rule ($d_{\text{pair}} < 0$ and capped size ratio ≥ 0.2). These are checks on the same generation-visible cases, not unseen tests.

C.4 COMPLETE ATTRIBUTION RESULTS

The main-table values were recomputed from the frozen per-query records, using the exact counts of candidates above and tied with the true source. All 229 benchmark queries have 20 distinct candidates and contain the true source exactly once. A Rank@1 of 0.050 can arise from a small number of unique wins or from uninformative 20-way ties; these are distinct outcomes. [Table 12](#) separates them.

The earlier 14-query SourcererCC and MSCCD pilot is different: every query has zero similarity to all 20 candidates. Their 0.050 forced Rank@1 is solely the expected credit under random tie-breaking; neither has a successful calibrated attribution. It is not evidence of useful source identification. Those pilot records also predate the stricter subprocess checks and are not used to validate the current tool installation.

Adaptation and decision scope. For clone-pair methods, attribution is a benchmark adaptation: detected regions become repository-level token-coverage scores, which rank the 20 candidates. This is not an attribution algorithm supplied by the original clone detector. Valid execution and readable inputs must be established before interpreting zero coverage as a detector outcome; unsupported languages and failed executions are not negative evidence. The reported calibrated credit tests the true source’s rank and reference band. The stored records do not contain calibrated decisions for every incorrect candidate, so they do not determine an operational abstention rate or false-attribution rate. These require candidate-specific calibration and a fully specified sampling-and-abstention rule.

Result-field semantics. In the frozen evaluator records, the stored attribution flag denotes band passage, not conjunctive attribution. The paper’s calibrated-attribution column is instead computed as top-rank credit times that flag. The released records expose band passage and calibrated attribution separately, and retain all 20 unrounded candidate scores for audit. A rounded similarity must never be used to reconstruct a near-threshold decision.

The AST-surrogate record labelled `treecen` is an AST-type-count cosine surrogate, not a faithful implementation of TreeCen. Its full 20-candidate score records reproduce the stored rank arithmetic, but 18% of queries contain at least one candidate with empty or otherwise invalid token evidence. Among queries without those invalid comparisons, weighted and query coverage yield different top-

Table 11: Detector performance across the four major rewriting axes. Same-language Java retains substantially more relative provenance signal than the other axes, but calibrated detection remains limited and often comes with increased false-positive rates. Cross-language rewriting amplifies provenance loss but is not necessary for it. Calibrated attribution (omitted from the columns) is zero except for CodeBERT on Java-to-Java (0.050) and k-gram on Java-to-JavaScript (0.004). TPR and calibrated attribution use exact stored decisions and tie counts; AUROC/FPR are the frozen per-axis summaries.

Detector	AUROC	TPR	FPR	Rank@1
Python to Python				
UniXcoder	0.635	0.000	0.000	0.250
CodeBERT	0.574	0.000	0.000	0.050
AST surrogate	0.487	0.150	0.277	0.000
TF-IDF	0.581	0.000	0.000	0.000
k-gram	0.415	0.000	0.000	0.000
Stat. profile	0.477	0.000	0.038	0.000
Token Jaccard	0.341	0.000	0.057	0.000
Python to JavaScript				
UniXcoder	0.625	0.000	0.000	0.250
CodeBERT	0.532	0.000	0.000	0.050
AST surrogate	0.586	0.000	0.013	0.050
TF-IDF	0.603	0.000	0.000	0.050
k-gram	0.447	0.000	0.000	0.000
Stat. profile	0.444	0.000	0.000	0.000
Token Jaccard	0.339	0.000	0.025	0.000
Java to Java				
UniXcoder	0.834	0.090	0.000	0.150
CodeBERT	0.705	0.372	0.132	0.050
AST surrogate	0.645	0.385	0.207	0.050
TF-IDF	0.708	0.244	0.033	0.050
k-gram	0.405	0.051	0.126	0.025
Stat. profile	0.539	0.077	0.081	0.000
Token Jaccard	0.251	0.179	0.237	0.000
Java to JavaScript				
UniXcoder	0.611	0.000	0.014	0.100
CodeBERT	0.558	0.030	0.002	0.000
AST surrogate	0.503	0.000	0.022	0.050
TF-IDF	0.538	0.000	0.008	0.000
k-gram	0.389	0.000	0.004	0.011
Stat. profile	0.429	0.000	0.000	0.000
Token Jaccard	0.219	0.000	0.081	0.000

Table 12: Attribution diagnostics. Unique wins, top ties, all-candidate ties and band passes are fractions of the 229 attribution queries; band passage alone is not successful attribution. Rank@1 and source-band attribution are means of fractional credits.

Method	Unique wins	Top ties	All-20 ties	Band passes	Rank@1	Band credit
UniXcoder	0.22	0.00	0.00	0.03	0.220	0.010
CodeBERT	0.06	0.00	0.00	0.11	0.060	0.010
AST-type-count surrogate	0.05	0.00	0.00	0.14	0.050	0.010
TF-IDF	0.05	0.00	0.00	0.04	0.050	0.000
k-gram	0.00	0.03	0.00	0.02	0.007	0.001
Statistical profile	0.00	0.00	0.00	0.02	0.000	0.000
Token Jaccard	0.00	0.00	0.00	0.06	0.000	0.000

Table 13: Input-view sensitivity for the encoder adaptations on 46 matched Java attribution queries from 19 source IDs. All queries retain the same 20 candidates. Accuracy counts unique correct top scores without a threshold; intervals are descriptive 95% source-ID bootstrap percentile intervals.

Adaptation and input	Correct	Accuracy	Interval
CodeBERT, repository view	6/46	0.130	[0.021, 0.280]
CodeBERT, Java source	4/46	0.087	[0.000, 0.200]
UniXcoder, repository view	14/46	0.304	[0.143, 0.478]
UniXcoder, Java source	19/46	0.413	[0.212, 0.622]

candidate sets in 72 of 82 cases. Both have unique winners on 30 queries; 20 of these winners differ. This sensitivity audit does not select an aggregation or validate the original detector.

Transport metadata audit. Repositories are transferred between machines, and transfer can introduce macOS AppleDouble sidecar files; 561 transferred rewrite directories contained them. The input views exclude a sidecar only when both its dot-underscore filename prefix and its AppleDouble magic bytes agree; genuine hidden source files are retained, and the broad repository-text loader rejects these non-UTF-8 files. Transport metadata is not benchmark source code. All 2,526 native input views are byte-verified against the canonical sources, covering 1,382 reference repositories and 1,144 distinct rewrite paths, and all native records reported in this paper use these verified views.

Encoder input-view sensitivity. The Java-source ablation uses the verified inputs. On 46 matched queries from 19 source IDs, CodeBERT identifies the unique top source in 4 cases with Java-source input and 6 with the repository-text view; UniXcoder does so in 19 and 14 cases, respectively (Table 13). Twelve of the 58 requested queries lack a complete 20-candidate Java-source view and are excluded explicitly. The direction differs across encoders and the intervals overlap, so no view is uniformly better. These are unthresholded, source-present ranking diagnostics, with no abstentions on the matched cohort; they do not establish calibrated attribution or determine which input view is primary. Intervals resample source IDs, whose fork independence has not been certified.

C.4.1 PAIRED SOURCE-ABSENCE DIAGNOSTIC

Table 14 reports computed decisions under frozen per-axis thresholds for 20 benchmark queries. For each source-present query we remove its true source, retain all 19 distractors, and apply the same threshold and unique-top prediction rule. Labels enter construction of this sensitivity condition and outcome assessment, never the prediction function. This is a 19-candidate paired removal experiment, not a new 20-candidate open-set benchmark or an independent population of unrelated queries. Thresholds are not refitted. The diagnostic covers 15 Java-to-Java and five Python-to-Python queries. It separates erroneous answers from abstention, but does not yet cover all 229 attribution queries.

Table 14: Source-present and paired source-absent decisions on 20 benchmark attribution queries. The decision rule is unique-top attribution with frozen per-axis thresholds; present-source counts use 20 candidates; the paired absent-source sensitivity removes the true source and retains 19 distractors without recalibration. Correct/wrong/abstained (C/W/A) and false attribution/ correct rejection (FA/CR) are query counts.

Method	Axis	n	C	W	A	FA	CR
CodeBERT	Java	15	0	5	10	5	10
CodeBERT	Python	5	0	0	5	0	5
UniXcoder	Java	15	0	1	14	1	14
UniXcoder	Python	5	1	0	4	1	4
SourcererCC	Java	2	0	0	2	0	2
SourcererCC	Python	1	0	0	1	0	1

C.5 LANGUAGE-SPECIFIC / JAVA-TIER RESULTS

C.5.1 PROVISIONAL JAVA-AXIS RECORD: STONEDETECTOR

An extended StoneDetector pairlist, separate from the clean-input row in [Table 1](#), covers the evaluated Java-to-Java Pair and Attr IDs in full, making it applicable-axis complete. Under the zero-on-error/unsupported missing-score policy it records AUROC 0.716 and best F1 0.515, with Pair TPR 16/78 and FPR 5/334, Rank@1 1/20 and calibrated attribution 0/20 under the conjunctive rule, alongside the same-axis comparison in [Table 11](#). This is an audit of recorded outputs, not certification of a validated baseline: unsupported comparisons and timeouts are stored as zero, so ranking and band construction need explicit missing-score handling. All ten errors are timeouts. Four are source-neighbor comparisons for `elasticsearch`, outside these evaluated cohorts. One is the `mvel-lombok` calibration comparison; three are the `mvel rewrite-lombok` negative pairs. The remaining two are Attr distractor comparisons (`commonmark-java` against `opensearch`, and `commons-jxpath` against `hadoop`). No evaluated positive comparison timed out. Removing queries with any timed-out positive, negative, calibration or candidate comparison leaves Pair TPR 16/75 and Attr 0/18. This exclusion is a sensitivity check, not an estimate of performance on the omitted queries.

Unsupported records affect at least one required comparison for 15 Pair queries and four Attr queries. Their zeros are not evidence of a valid no-match result. The raw Pair false-positive count is 5/334 under this policy. Neither this provisional record nor the timeout exclusion establishes correct parsing, local-copy recall, or robustness to the whole-repository token denominator. Applicable-input self-copy and partial-copy checks, explicit unsupported/error outcomes, and rerunning affected comparisons are needed before treating this record as a validated baseline.

D EXPLANATORY ANALYSES

D.1 WHAT PROVENANCE SIGNALS SURVIVE REWRITING?

D.1.1 WHAT THE SURVIVING LEXICAL SIMILARITY IS MADE OF

Design. This diagnostic splits token-frequency cosine between a source and artifact into contributions from contract vocabulary and other tokens. It uses term-frequency weighting without IDF, so the additive split is exact for this diagnostic, not for the primary TF-IDF detector. A token is assigned to the contract component when it occurs in the frozen contract material of that instance (the purpose statement, the benchmark test cases, or the golden outputs) and to the incidental component otherwise; the split is computed per instance and the two components sum to the full cosine with residual $\leq 5.6 \times 10^{-17}$ over the evaluated instances on the benchmark’s four rewriting axes. The token assignment uses no learned model or fitted parameter.

Results. The separately summarized component medians are nearly equal (0.143 contract and 0.143 incidental), while the median per-instance contract share is 0.530. The contract component is

Table 15: Contract share of token-frequency cosine, by axis. The share is the fraction of the normalized dot product contributed by tokens occurring in the instance’s frozen contract material; the two components sum to the full cosine exactly.

Axis	Median contract share	IQR	Median diagnostic cosine
py→js	0.67	[0.56, 0.77]	0.17
java→js	0.57	[0.51, 0.63]	0.27
java→java	0.48	[0.38, 0.56]	0.44
py→py	0.40	[0.26, 0.47]	0.42

larger on 55.6% of instances. The share is not uniform across the benchmark’s axes: it is highest where source and artifact are written in different languages (py→js 0.67, java→js 0.57) and lowest on the same-language axes (py→py 0.40, java→java 0.48), the ordering expected if cross-language rewrites retain contract vocabulary while discarding incidental identifiers along with the syntax (Table 15).

The split also provides a diagnostic of the encoder margin. The contract share is *negatively* associated with the UniXcoder margin over the same instances (Spearman $\rho = -0.227$, 95% CI $[-0.327, -0.117]$; Figure 11b plots the two quantities per instance over the four axes): higher contract share is associated with a lower UniXcoder margin. The diagnostic’s full cosine correlates with the frozen TF-IDF score at $\rho = 0.353$, further distinguishing their numerical scales. These associations are consistent with functional vocabulary contributing to residual similarity, but do not identify which features the encoder uses or establish a causal mechanism.

D.1.2 DETERMINISTIC CHANNELS PREDICT FAILURE WITHOUT PROVIDING DETECTION

Design. The study was pre-registered and the analysis plan sealed before computation. The three deterministic diagnostic channels (identifier, string-literal, import-call overlap, extracted with tree-sitter) are summarized per instance by their channel margins (score minus per-source band). A logistic probe on the three margins is trained to predict which benchmark artifacts pass a detector’s band, with folds fixed to the released leave-one-source-group-out audit splits; no hyperparameter is tuned on any test fold. The pre-registered folds cover 37 source groups (154 instances); the remaining sources lie outside every fold and are excluded from both training and test. Decisions use the same grouped-band rule as the benchmark: the pair (A, B) is called detected when the probe score exceeds the maximum probe score over A ’s same-domain negatives. Leave-one-channel-out refits report which channel carries the prediction.

Results. The margins alone predict band passage. For the lexical (tfidf) target, which has 12/154 positives, the probe reaches held-out AUROC 0.80; dropping the identifier channel costs the most (0.70), dropping string literals 0.74, dropping import calls nothing (0.80). For the encoder (UniXcoder) target, with 4/154 positives, AUROC is 0.75 and flat under every leave-one-channel-out refit, consistent with the encoder signal not being concentrated in any one deterministic channel. The grouped-band decisions tell the other half: the probe recovers only 15/154 (lexical) and 13/154 (encoder) instances, barely above the detectors’ own 12 and 4. Predicting a detector’s band outcome is different from establishing provenance. The fitted probe recovers some additional positives, but these counts alone do not establish source-specific detection without its false-positive context. In particular, without the identifier channel the grouped-band recovery *rises* to 31/154: identifier margins are also high for same-domain negatives. This is consistent with shared vocabulary helping predict detector behavior without necessarily improving a calibrated provenance decision.

D.2 STRUCTURAL AND RUNG-WISE REWRITE EVIDENCE

Structural retention. The frozen descriptor study records source-to-artifact structural similarity for 1,156 source-artifact pairs spanning the rewriting rungs. Median retention is 0.156 for file count, 0.011 for repository size, 0.141 for AST-trigram cosine (400 pairs with complete parses), 0.008 for identifier Jaccard, 0.005 for string-literal Jaccard, and 0.003 for import/call similarity (Figure 5a). These descriptors record broad implementation changes, but aggregate similarities do not establish

that every artifact changes every channel. A file-count ratio measures the number of files, not the fraction of original files retained; import/call similarity is not a dependency-DAG identity test.

Rung-wise decay. Restricting to fully conforming artifacts ($n = 101, 85,$ and 79 across the full-rewrite, contract, and critic rungs), the embedding detectors are already usually below the source band at the mildest rung. The median UniXcoder margin is -0.19 , and its band-pass rate is $1.0\text{--}3.5\%$ across rungs. Only the lexical TF-IDF channel retains a gradient. Its band-pass rate falls from 0.376 to 0.341 to 0.203 across the three rungs (Figure 5b). The critic rung feeds surviving source names back to the agent. Its accepted artifacts have lower lexical band passage, but the rung populations differ, so this comparison does not isolate the critic’s effect.

Generator validity and evasion. Under the identical pipeline, the valid rewrite rate rises from 0.875 (GLM-5.3) through 0.925 (DeepSeek-V4.1-Flash) to 0.950 (GPT-5.6-Sol), while evasion among valid artifacts stays between 0.83 and 0.95 (Figure 5c). Section D.6 reports the per-model records.

D.3 REWRITE STRENGTH AND DETECTION DIFFICULTY

Design. One round of the agentic pipeline is already a full behavior-constrained reconstruction: the repository is decomposed into a specification and units, each unit is rewritten in isolation, and the units are composed against the source’s conformance oracle. Chaining three rounds, where every round re-decomposes the repository produced by the previous round while the oracle stays fixed to the original source, gives a controlled ladder of rewrite depth: behavior is pinned to the same tests while the artifact is pushed further away from the source’s surface. We run three rounds on the 40 Python sources of the agentic pool with the same rewriting model as the main agentic runs, score every accepted round (conformance ≥ 0.99) with the frozen detector suite and the deterministic channel margins, and ask a single question: does provenance evidence collapse at the first reconstruction, or does it keep decaying while behavior is preserved? The analysis is fixed in advance: median margins per depth, and paired Wilcoxon signed-rank tests between consecutive depths over chains complete at both, with saturation claimed only when the drop from depth 1 to 2 is significant and the drop from depth 2 to 3 is not.

Results. Behavior is pinned: median conformance is 1.0 at every depth. Median detector scores are already below the reference bands after the first reconstruction (Figure 11a). Every detector’s median margin is negative at depth 1 (tfidf -0.090 , token Jaccard -0.107 , UniXcoder -0.138), and at most 6 of 25 sources still clear their band. The paired tests find an initial decline without a statistically resolved additional decline: over chains complete at consecutive depths, the drop from depth 1 to 2 is significant for tfidf ($p = 0.020$), token Jaccard ($p = 0.009$), and UniXcoder ($p = 0.041$), while the drop from depth 2

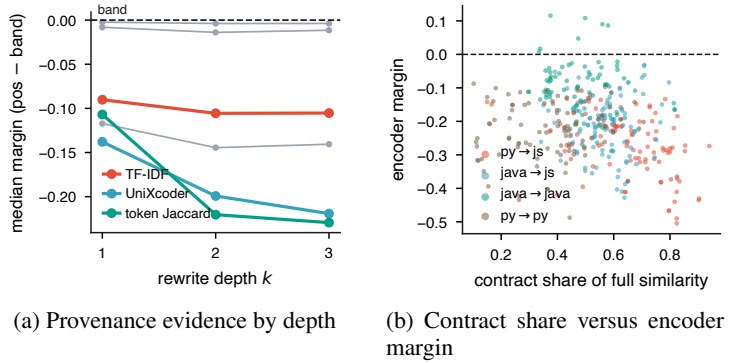


Figure 11: (a) Median detector margin (positive score minus the per-source band) over rewrite depth for three focal detectors, the remaining lexical and encoder detectors in grey (the stat-profile channel lies below -5 at all depths and is omitted from the panel), and the band as the dashed line. (b) Contract share of the lexical cosine against the UniXcoder margin, one point per instance on the benchmark’s four rewriting axes (Section D.1.1).

to 3 is not ($p \geq 0.075$), the prespecified pattern used as a saturation diagnostic. Non-significance of the later change does not establish equivalence or a true plateau. The deterministic channels echo the same shape: string-literal margins fall from -0.073 to -0.138 between the first two rounds and

Table 16: Within-source paired comparison of same-language versus cross-language artifacts on 158 dual-axis sources. Margins are encoder margins against the source-specific band (medians); Δ is the per-source same-language minus cross-language margin.

Pair	n	Same	Cross	Δ median	Wilcoxon p
Python (132 sources)	132	-0.151	-0.175	+0.024	1.2e-6
Java (26 sources)	26	-0.114	-0.160	+0.042	2.3e-5

then hold at -0.139 , and identifier margins drift without recovering (-0.174 , -0.255 , -0.235). Of the 40 chains, one stops at round 1 when composition cannot recover conformance; 23 complete all three rounds, and 25, 24, and 23 sources contribute an accepted artifact at depths 1, 2, and 3. Low margins after the first observed reconstruction support early loss of detector-observable evidence. Later accepted rounds preserve the tested behavior; failed and incomplete chains limit conclusions about the cost of repeated rewriting.

D.4 SAME-LANGUAGE VS. CROSS-LANGUAGE PROVENANCE

Design. Cross-language rewriting should remove more provenance evidence than same-language rewriting, because language-specific lexical and structural traces may change substantially under translation. We test this within sources: 158 release sources carry a verified artifact on both their same-language and their cross-language axis (132 Python, 26 Java), giving a paired comparison that holds the source fixed. For each source and detector we take the same-language margin minus the cross-language margin against the source-specific band, with Wilcoxon signed-rank tests pre-registered before computation.

Same-language rewriting retains a modest encoder-margin advantage. Encoder margins are higher on the same-language artifact for 74% of paired sources (Table 16). The median advantage is $+0.024$ for Python pairs (Wilcoxon $p = 1.2 \times 10^{-6}$) and $+0.042$ for Java pairs ($p = 2.3 \times 10^{-5}$). TF-IDF shows no statistically resolved advantage ($p \geq 0.27$), so the encoder advantage is not mirrored by that lexical measure. This does not identify the representation features responsible. Median encoder margins remain below the band in both settings. The corresponding median uncapped source-to-artifact size ratios are 0.010 (py→py) versus 0.012 (py→js) and 0.017 (java→java) versus 0.013 (java→js) on the paired sources.

Reading. Cross-language rewriting has lower encoder margins in the within-source comparison, consistent with the axis-level pattern in Section 3.1. Holding sources fixed reduces composition effects, but does not isolate translation from all other implementation changes.

D.5 REPOSITORY SCALE AND REWRITE SCOPE

Design. Two pre-registered questions, sealed before computation: does source complexity explain where detection fails, and does restricting the source to the slice its oracle actually covers recover encoder signal? Complexity measures (LOC, file count, module count, import-graph edges, public-API count) are computed statically from the pinned source clones. The oracle-coverage slice of a source is the set of files touched by its conformance harness plus their transitive imports; the definition was fixed before any detector was run on the slice view.

Terciles. Sources are split into complexity terciles within each major axis (the source, not the instance, is the tercile unit), and per-detector median margins and calibrated TPR are reported per tercile with source-cluster bootstrap intervals. The result is flat: UniXcoder’s median margin stays between -0.28 and -0.11 in every tercile of every measure on every major axis. The margin is slightly less negative for larger sources on the Python axes (e.g. LOC terciles on py→py: -0.279 low vs. -0.230 high) and essentially unchanged on java→java, where calibrated TPR never exceeds 0.15 in any tercile and is 0 everywhere outside java→java. The gap persists across these complexity strata, so source complexity alone does not explain the observed low calibrated detection.

Slice control. Re-scoring UniXcoder with the source restricted to its oracle-coverage slice (artifact unchanged, 88 benchmark instances with file-anchored slices) moves the positive score by a median of -0.033 , so restriction slightly *lowers* similarity, and only 5/88 instances cross the band that 0/88 cross with the whole repository. This control gives little support to source-file dilution as the sole explanation. It does not exclude all effects of repository aggregation or validate the slice as a complete semantic boundary.

D.6 GENERATOR-BACKBONE ABLATION

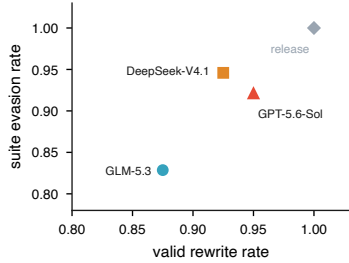


Figure 12: **Generator-backbone ablation** under a fixed agentic rewriting pipeline on 40 Python sources. We compare the default GPT-5.6-Sol backbone with GLM-5.3 and DeepSeek-V4.1-Flash, varying only the generator backbone. The x -axis reports the fraction of sources reaching full behavioral conformance; the y -axis reports, among valid rewrites, the fraction below the source-specific reference band on all seven static channels. Released rewrites of the same sources (grey diamond) provide the reference configuration; their validity is ensured by release selection.

Generator backbone. GPT-5.6-Sol is the default generator backbone. We additionally evaluate GLM-5.3 and DeepSeek-V4.1-Flash while holding the agentic rewriting pipeline (Section A.2), the 40 Python sources, and the evaluation protocol fixed. GLM-5.3, DeepSeek-V4.1-Flash, and GPT-5.6-Sol produce valid rewrite counts of 35/40, 37/40, and 38/40. Among these valid outputs, 29/35, 35/37, and 35/38 fall below the source band on all seven static channels, against 40/40 for the release rung over the same sources (Figure 12), with wide margins (median UniXcoder gap -0.81 versus -0.27 for release) and no embedding detector catching any valid artifact. Frequent evasion of the seven static channels is therefore not specific to a single tested generator backbone. This comparison does not establish generator independence or a causal ordering by capability.

D.7 MULTI-SOURCE PROVENANCE

The release benchmark covers single-source artifacts. As a first probe of mixed-origin code, we construct composite artifacts by merging two verified same-language rewrites of the full-rewrite rung, each parent under its own top-level module directory. Pairing is deterministic over hash-sorted sources, with mutual same-domain neighbours excluded, giving 17 Python and 14 Java pairs. Each composite is scored against both parents with the frozen lexical detectors on the canonical corpus, after re-verifying all 189 involved frozen source bands exactly. Same-language non-parent sources serve as specificity controls.

Mixing dilutes per-parent signal. Under TF-IDF, the channel with the strongest single-source margins, the mean margin drop from the single-source reference to the composite is 0.119, against a median source band of 0.173. Parents above their own band fall from 25/62 for the single-source rewrites to 7/62 for the composites, and both parents clear their bands simultaneously in only 1/31 pairs. The effect is strongest in Python, where none of the parent scores remains above its band (12/34 parents above band before, 0/34 after); in Java, 7/28 parents remain above band, consistent with the stronger same-language Java signal of Table 11. The k -gram and statistical profile channels already sit below band for single-source rewrites and remain there. False positives remain: 6/155 non-parent controls exceed their bands.

These file-level composites are harder to associate with both parents under the evaluated whole-artifact scores. The result motivates testing mixture-aware alternatives such as segment-level matching; it does not establish that provenance signals can never compose. We leave spec-merged blends and mixture-aware detectors to future work.

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E CASE STUDIES

This section makes the aggregate results legible through concrete instances: what a rewrite that evades the seven static channels looks like, which repositories resist evasion, and where the con-

Table 17: Three evading release artifacts (B5x critic rung): positive score (band) per detector. All seven margins are negative on their native detector scales.

Detector	jwcrypto (py→py)	dask (py→js)	flink (java→js)
tfidf	0.188 (0.291)	0.029 (0.203)	0.017 (0.103)
<i>k</i> -gram	0.070 (0.203)	0.000 (0.297)	0.023 (0.141)
token Jaccard	0.086 (0.320)	0.002 (0.390)	0.000 (0.022)
profile	−9.57 (−5.28)	−26.6 (−3.5)	−27.2 (−4.7)
UniXcoder	0.814 (0.948)	0.617 (0.857)	0.506 (0.861)
CodeBERT	0.9956 (0.9958)	0.989 (0.996)	0.987 (0.998)
GraphCodeBERT	0.981 (0.983)	0.942 (0.979)	0.955 (0.988)

struction itself fails. Every number comes from the frozen per-instance records released with the benchmark and from the study materials cited inline.

E.1 THE COMPLETE AGENTIC REWRITING BACKEND

The backend that re-executes the full pipeline below instantiates [Algorithm 1](#): every stage is a separate headless coding-agent session with no shared context, driven by a resumable scheduler that runs many pipelines in parallel.

Stages. A *decompose* session reads the source repository and writes a unit inventory: three to eight core modules ordered bottom-up, each specified by a responsibility, a precise statement of observable behavior, the interface it must expose, and the literal data strings its behavior fixes. File names, directory layout, and the original project’s identifiers are forbidden in the inventory. Each unit is then rewritten by a *fresh session in its own directory* that receives only that unit’s specification and the interfaces of the units it depends on, never the source and never a sibling implementation. A *compose* session assembles the units bottom-up, resolves interface mismatches by adjusting callers, adds the conformance entry point, and iterates against the behavioral oracle until conformance reaches 1.0: the candidate is copied to the evaluation host and run against the source’s captured golden cases at every iteration. The composed repository then becomes the input to the next cycle: the decomposition is re-derived from scratch on the rewritten code rather than inherited, so every cycle performs a fresh reconstruction while retaining the same provenance path. A final *critic* session receives the source-chosen tokens that survive in the composed repository and removes them by renaming identifiers and rewording comments and messages, gated on conformance staying 1.0. Values the specification marks as externally required (protocol field names, file-format keys) are kept: behavior wins over leak removal. The exact stage prompts are reproduced in [Appendix A.5](#).

Models. The same driver runs the identical pipeline with the default GPT-5.6-Sol backbone and, for the backbone ablation, GLM-5.3 and DeepSeek-V4.1-Flash against the same oracle. The jwcrypto trace in [Appendix E.2](#) is one full execution: six units decomposed, rewritten in isolation, and composed to conformance 1.0; a second cycle over the composed repository; a critic pass; conformance 1.0 (strict) throughout.

E.2 WHAT A SUCCESSFUL EVASION LOOKS LIKE

The aggregate pattern. Of the 1,128 instances in the frozen release suite, 87% fall below the band of all seven channels in this diagnostic: four lexical/statistical channels and three encoders. This does not include the AST surrogate or native clone tools. Evasion is not confined to obscure repositories: every py→py and py→js instance evades all seven detectors, as do 96% of java→js instances. Detection lives almost entirely on the java→java axis, which supplies 87% of the detected instances ([Appendix C.5](#)).

Three cases. [Table 17](#) makes the difference between similarity and evidence concrete. Each source–rewrite score is below its detector-specific band, even when absolute similarity is high. For jwcrypto, CodeBERT scores the rewrite at 0.99560 against a band of 0.99580, a margin of only

2×10^{-4} . The example illustrates a threshold-dependent failure to distinguish derivation from the reference code, not the disappearance of similarity. Native score scales differ, so the numerical margins should not be ranked across detector families.

Which repositories resist evasion. The repositories with at least one detected instance are overwhelmingly small java→java utility libraries with ecosystem-standard vocabulary (the Apache Commons family, `joda-time`, `snakeyaml`, `picocli`, `google-java-format`). These examples retain domain vocabulary in public interfaces, including names such as `Duration` or `OptionParser`. Such vocabulary is a plausible source of surviving similarity, but the cases do not identify which tokens determine encoder scores. GraphCodeBERT contributes to 82% of the detections and CodeBERT to 58%; `univocity-parsers` is the most detected artifact in the benchmark (six of seven detectors).

Required behavior can retain recognizable evidence. On `jwccrypto`, whose release artifact sits below band on all seven static channels, re-executing the full backend of [Appendix E.1](#) on the same source shrinks the surviving source vocabulary while behavior stays intact: two complete cycles plus a critic pass cut surviving source-chosen tokens from 143 to 45 while conformance stays 1.0. The surviving vocabulary is the JOSE term set (JWK/JWS/JWE field names fixed by RFC 7515/7516) that the conformance cases compare against: the tested outputs constrain those strings. Required behavior can thus retain recognizable evidence despite repeated reconstruction.

E.3 WHEN LEXICAL EVIDENCE FAILS

The lexical/statistical channels sometimes complement the encoders, but their evidence is limited in these case-study records. Only 15% of the detected instances are caught by a classical detector that every encoder misses, and all eight are java→java (`java-jwt` and `joda-time` by `tfidf`, `typetools` by `tfidf` and token Jaccard, one `commons-digester` artifact by `k-gram` alone). The reverse asymmetry is larger: `commons-text` (java→java) is caught by CodeBERT, GraphCodeBERT, `tfidf` and `UniXcoder` while its `k-gram` positive is 0.031 against a band of 0.086 and its token Jaccard 0.010 against 0.057. Once a rewrite renames identifiers and reorders tokens, `n-gram` overlap falls below the floor that unrelated repositories sharing common library boilerplate already establish. These cases concern lexical/statistical similarities, not the provisional SourcererCC, MSCCD, StoneDetector, TACC, or Oreo evaluations. They do not establish dominance over native clone detectors.

E.4 WHEN AN LLM / CODING AGENT FAILS

The source-visible camouflage recipe often fails behavioral checks. In the independent reimplementation ladder, the arm whose agent *reads the full source* and is asked to camouflage it in place produces artifacts that often fail the behavioral gate: 14/29 do not execute at all, and the best conformance among those that run is 0.43 (`django`, `py→js`); `commonmark-java` runs at 0.0. The arm that sees only the behavioral contract reaches conformance 1.0 on 24/29 sources. These outcomes compare generation recipes with different inputs and constraints; they do not show that access to source code itself causes behavioral failure.

The tested mechanical interventions give limited valid evasion. Five mechanical interventions applied to 20 release artifacts: identifier splitting, merging, and string-literal renaming each destroy behavior in 20/20 artifacts; comment laundering preserves behavior in 16/20 but moves the `tfidf` and `k-gram` margins by at most 0.009; comment stripping preserves behavior in 20/20 with margin change exactly 0. The only margin that moves substantially (+0.69) is the profile channel, the one channel the construction already controls. The tested primitives either break behavior or leave these lexical margins nearly unchanged. This does not rule out other behavior-preserving obfuscation methods.

Single-shot evasion varies by language axis. The strongest single-shot generator (`gpt-5.6-sol`, reimplementation arm) produces artifacts that evade every detector on 15/21 sources, but on java→java only 1/6 evade; four of the five caught java→java artifacts are caught by CodeBERT and GraphCodeBERT. The Java cases retain more detectable signal in this study,

consistent with the examples in [Appendix E.2](#). This does not establish a general relationship between generator capability and evasion.

E.5 MULTI-SOURCE PROVENANCE

Every artifact in the frozen benchmark descends from exactly one source repository; multi-ancestor artifacts (blends of two sources, or a rewrite seeded from a competitor’s implementation) are outside the current construction. The pair task’s negative sets already probe the boundary: each source is scored against its same-domain competitors (e.g. `jwtcrypto` against `joserfc`, `pyjwt`, `pyjwkest`, `jwskate`), so a detector must separate derivation from same-domain independent implementation. A first small-scale probe that merges two verified rewrites at the file level finds that per-parent provenance signal is diluted below the source bands, so both parents of a composite are rarely recovered together ([Appendix D.7](#)). Constructing blended artifacts by decomposing two sources and merging their specifications remains a future extension of the pipeline, distinct from this file-level composition probe.

E.6 FAILURE CASES OF BENCHMARK CONSTRUCTION

Purpose-only specifications under-determine behavior. The ladder arm whose agent receives only a one-paragraph purpose statement produces no artifact that reaches the 0.99 conformance gate (0/29); five do not execute, and the conformances that run span 0.0–0.5. A purpose statement fixes what a system is *for* but not what it *does*; a purpose statement alone is insufficient under this study’s strict gate.

Contracts can under-determine behavior too. Even the interface-only arm fails on sources whose observable behavior is not captured by the extracted contract: `commons-digester` reaches only 0.375, `commons-dbutils` 0.82, `jsoup` 0.85. The oracle, not the contract, is the ground truth; instances whose golden exercise behavior outside the contract are exactly where independent reimplementations fall short, and they are excluded by the study’s strict verification gate. This is distinct from the release criterion, which permits partial conformance.

Purpose descriptions leak generic vocabulary. The purpose statements carry 0–45 source-chosen tokens per source, but inspection shows they are generic domain words (`admin`, `manage`, `preview`) rather than identifying names; we disclose the counts rather than claim zero leakage.

Interventions can backfire. The comment-laundering intervention of [Appendix E.4](#) moves the profile margin by +0.69. Under the score-minus-band convention, a positive change moves toward detection, not away from it. The profile channel is sensitive to the altered implementation statistics; this intervention does not establish that those statistics uniquely identify provenance.

F ROBUSTNESS AND LIMITATIONS

F.1 METRICS, CALIBRATION, AND STATISTICAL UNITS

Independent units. Pairwise rows are repository-pair observations clustered by generating source; attribution rows are query repositories with fixed candidate sets. Rewrites of one source are not treated as independent source replicates. Pooled descriptive metrics use all available instances, while uncertainty checks resample sources and retain all their associated instances. All intervals are conditional on the frozen sources, generated artifacts, negative neighborhoods, and task manifests; they do not represent uncertainty over a new generation run.

Pairwise and attribution metrics. AUROC and best F1 summarize threshold-free ordering and the best retrospective threshold, respectively; best F1 is not a deployable operating point. For detector d and source s , the independent-code band is $\beta_d(s) = \max_{n \in \mathcal{N}(s)} S_d(s, n)$. Calibrated TPR and FPR apply the strict test $S_d(q, c) > \beta_d(s)$ to positive and same-domain negative pairs. Rank@1 gives fractional credit when the true source is tied at the top. Calibrated attribution additionally

Table 18: What each detector family measures, and the failure exposed by the benchmark. This table interprets, rather than repeats, the numerical suite in Table 1.

Family	Evidence used	Observed failure mode
Lexical/statistical	Tokens, character substrings, repository shape	Reconstruction reduces incidental overlap; shared contract vocabulary and style can also raise non-source scores
Token clone	Indexed file/block overlap	Near-random discrimination on readable inputs, with incomplete repository and language coverage
Structural/learned clone	Parsed functions, AST regions, metrics, pair classifiers	Provisional native results require parsing, mapping, and band validation; the AST surrogate has similar pooled TPR and FPR
Repository embedding	Pooled pretrained window representations	Relative ordering survives more often than calibrated attribution; evidence varies by language axis

Table 19: AST-surrogate calibration sensitivity on the same eligible queries in both columns. Pair contains 417 queries from 111 sources and 2,037 negative pairs; Attr contains 96 queries from 80 sources. Attr uses conjunctive top-tie credit. No detector was rerun and no benchmark instance was changed.

Outcome	Original band	Single-deletion minimum
Pair true positives	46/417	71/417
Pair false positives	255/2,037	354/2,037
Attr success credit	1/96	2/96

requires that top source to clear its band; band passage alone is not attribution. Missing parses, unsupported languages, timeouts, and empty eligible views remain unavailable outcomes rather than numerical zeroes.

Table 18 connects those metric definitions to the distinct applicability and decision failures observed for each detector family.

Uncertainty and multiplicity. The primary benchmark reports descriptive task metrics rather than treating every detector–axis cell as an independent hypothesis test. For calibrated success rates, 4,000 source-cluster bootstrap resamples provide percentile 95% intervals; for example, Pair TPR is 0.023 for UniXcoder ([0.004, 0.048]) and 0.074 for CodeBERT ([0.035, 0.120]). Targeted paired studies define the source as the unit, report the paired effect and source bootstrap interval, and use two-sided Wilcoxon tests only for their sealed primary comparison. The full per-axis record, control tier, strict-positive sensitivity, and native-tool applicability audit follow below.

F.1.1 SENSITIVITY TO CALIBRATION NEIGHBORS AND MISSING COMPARISONS

Single-neighbor deletion. We use the AST surrogate’s unrounded pair scores to perturb only the calibration neighborhood. For a source with at least two neighbors, define

$$\beta_d^-(R_s) = \min_{N \in \mathcal{N}(R_s)} \max_{M \in \mathcal{N}(R_s) \setminus \{N\}} d(R_s, M).$$

This is the smallest band obtainable by deleting one neighbor, equivalently the second-largest source–neighbor score, counting repeated maxima. It measures the strongest single-deletion effect, not an average over deletions. Artifacts, task membership, negative pairs and attribution candidates remain fixed; the deleted calibration neighbor remains in the evaluation set. The deletion is determined by source–neighbor scores, not query outcomes. We exclude 17 Pair and four Attr queries with singleton neighborhoods from both columns, since deletion leaves their band undefined.

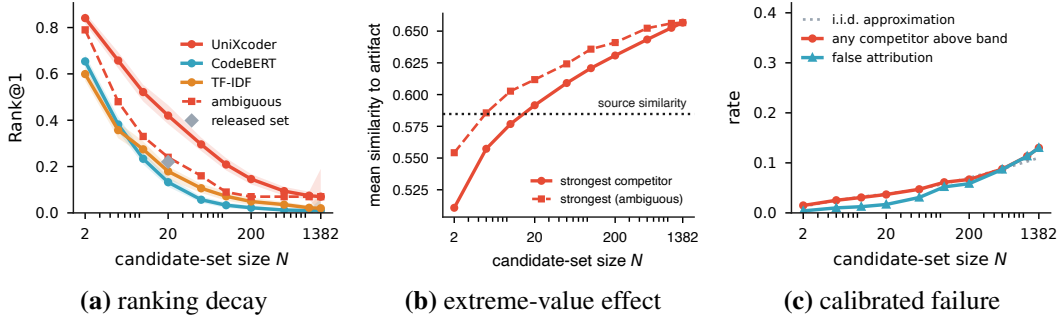


Figure 13: Candidate-pool sensitivity with candidate-set size N on a log scale. (a) Tie-aware Rank@1 decays smoothly as N grows, both for random competitor draws (solid) and for domain-ambiguous draws that keep the $N - 1$ pool repositories most similar to the source (dashed, strongest detector). The grey diamond marks the released 20-candidate task. (b) The extreme-value mechanism behind the decay: the mean similarity of the strongest independent competitor grows with N while residual artifact-to-source similarity (dotted) is constant in N ; the two cross already near $N \approx 20$. (c) The empirical rate of drawing at least one above-band competitor is shown alongside the independent-draw approximation $1 - (1 - p)^{N-1}$, where p is the per-query fraction of pool competitors above the frozen source band. This expression is not a union bound. The false-attribution curve uses the study’s fixed-band rule and does not establish deployment error control.

The band decreases for 404 Pair and 91 Attr queries. More positives cross the lower band, but so do more negatives (Table 19). This sensitivity does not validate the neighbors’ independent origin or validate published TreeCen or provide a candidate-available prediction rule: it retains the generating source’s grouped band. The other six channels’ rounded per-instance summaries are not used to reconstruct this perturbation. The frozen encoder records behind Table 1 store each source’s band but not its individual source–neighbor scores, so this sensitivity has not been computed for the repository encoders. A source-isolated calibration and source-absent evaluation remain needed.

StoneDetector records without flagged missing comparisons. Extending the timeout-only exclusion in Appendix C.5.1, we exclude a query if any required positive, negative/candidate or source–neighbor comparison is marked error or unsupported. This leaves 60 Java-to-Java Pair queries from 22 sources, with TPR 10/60 and FPR 5/248, and 14 Attr queries from 14 sources, with conjunctive success 0/14. These are conditional summaries of records without those flags, not performance estimates for omitted queries. Absence of a flag does not certify parsing or a valid no-match result; self-copy, partial-copy and local-coverage checks remain unperformed. The provisional record is the one of Appendix C.5.1. The released audit materials record the input hashes and a read-only standard-library reproduction of both analyses.

F.2 NEGATIVE/CANDIDATE-POOL SENSITIVITY

Larger candidate pools create stronger competing matches. The primary task fixes 20 candidates per query. In the pool-size sensitivity, UniXcoder’s Rank@1 under random competitor draws falls from 0.84 at $N = 2$ through 0.42 at $N = 20$ to 0.07 over the full 1,382-repository pool (Figure 13a). This does not replace the primary Rank@1 of 0.220: the random draws and the fixed benchmark candidate sets differ. Source similarity is fixed while the strongest competitor score increases with pool size (Figure 13b), explaining how retrieval can worsen without a change in the rewrite itself.

The chance of drawing an above-band competitor also grows (Figure 13c). The dotted curve is an independent-draw approximation; a general union bound would instead be $\min\{1, (N - 1)p\}$. The fixed source band does not imply a constant operational abstention rate, because that rate also depends on candidate scores and the prediction rule. These curves stress-test candidate competition, rather than validate open-set attribution or query-level error control. The fixed 20-candidate task is harder than random draws at the same size, consistent with the role of its same-domain distractors.

F.3 ADDITIONAL VALIDITY AUDITS

Does the acceptance gate test behavior or formatting? The behavioral oracle compares program outputs, so acceptance could in principle hinge on superficial formatting rather than computation. We audit this on the generator-frontier study of [Section D.6](#) by classifying every conformance case (1,011 cases over 40 sources) into core cases, which check computational behavior that the specification requires, and surface cases, which check formatting, persona, or greeting-style output. A strong model performs the classification and every doubtful case defaults to core. The suite is almost purely behavioral: 1,005 cases are core and only 6 are surface, all in two sources whose contracts involve conversational or human-facing output (ChatGLM-6B and TagStudio).

Relaxed-gate sensitivity. We then re-gate the study rewrites with a two-tier oracle that keeps exact comparison on core cases but checks only exit status, error class, and output type structure on surface cases. All 37 valid rewrites in the evaluated generator arm still pass at conformance 1.0. This holds even after an agent pass that rewrites the surface outputs themselves: persona and branding strings disappear, the count of surviving source-chosen tokens drops from 740 to 539, and core behavior is untouched. The longest core-case outputs are computational renderings such as markup escaping, table layout, and changelogs, not stylistic prose.

Reading. Core computational cases dominate this generator study. Relaxing the six surface cases preserves its validity outcomes. This supports the behavioral role of the oracle in this study, without establishing the same case composition for every benchmark source.

F.4 LIMITATIONS

The behavioral oracle checks a finite set of deterministic behaviors visible to the generator, so passing it establishes oracle-covered behavior, not whole-repository equivalence; the benchmark contract allows partial positive-case conformance. The bounded unseen-input audit of [Section B.3](#) uses benchmark instances. Its inputs were authored by this project’s own agent rather than an independent party, and two original-side environment repairs were applied before execution. The audit therefore covers the stated contract rather than whole repositories. The adapter token limit excludes degenerate artifacts but does not prove that a thin adapter performs only interface mapping. Oracle and environment requirements also determine which repository behaviors can be evaluated.